



Meena Needs a Computer Now: Co-design Solutions to Barriers using Fictional Inquiry for Women in Computing

Exploring the Tertiary Education Space in Bangladesh

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CCS CONCEPTS • Human-centered computing • Human computer interaction (HCI) • Empirical studies in HCI

Additional Keywords and Phrases: Women in Computing, stereotype barriers, ICTD, Qualitative-NLP analysis.

Gender disparity in computing is a well-explored concern. To complement existing research, we use a non-intrusive exploration method of Fictional Inquiry. Followed by co-design workshop to explore solution approaches to increase women's participation in computing. The study involved n=48 women in computing (WiC) students in computing from 4 universities across Bangladesh. Participants shared their challenges and desired solutions which we analyzed using qualitative methods along with Natural Language Processing based tools to illustrate the problems from various perspectives while minimizing potential biases. The research team reflects on methodological decisions, policy implications,

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ACM 1557-7325/2025/8-ART

<https://doi.org/10.1145/3762808>

and the need for culturally and contextually appropriate tools and datasets to better support WiC research in resource-limited settings.

1 INTRODUCTION

The title of this paper refers to Meena, a female cartoon character designed to break gender stereotypes in South Asia. In the early 2000s, Meena's stories raised awareness about the importance of girls' education. Many young girls inspired by Meena are now female students in computing at the tertiary level in 2023. They are now considering effective ways to address societal barriers to pave their pathway for women in computing (WiC). The societal barriers resulting in underrepresentation of women in computing is a long-standing problem worldwide and has been explored in numerous prior works [25, 26, 65-67, 70]. According to Clark, there may be a "leaky pipeline" of women as they drop out at various phases of tertiary education toward final career goals [65]. Despite various initiatives by the government and companies to achieve gender parity in computing, the representation of women remains low even in developed regions [33, 67, 70]. It is present in economically stable countries or countries that have addressed gender neutrality very seriously. For example, the number of WiC has been steadily declining over the years in USA over past few decades [105, 106] and the number is low in a gender-neutral country like Norway [107]. This problem is more acute for the marginalized women in South Asia who face significantly larger barriers such as sociocultural norms, influences, and infrastructural issues because of limited resources [26, 27, 60]. WiC research in Asia has focused on the socio-cultural context primarily exploring the challenges women face while entering a computing career showing strong regional and cultural influence [25-27, 31-35]. These research brought forth the existing challenges and various practices to address them to increase WiC representation [25, 26, 31-35, 65-67, 70, 105-109]. This work complements existing research by exploring how Women in Computing (WiC) perceive barriers when expressing themselves freely through letter writing, without fear of judgment. It examines the intensity of their feelings and provides design options to address these barriers. Using innovative methods and analysis, this research investigates the desired solutions and the intensity of perceived barriers among tertiary-level computing students in a developing country.

Research works in Bangladesh have revolved around policy and infrastructural suggestions provided by the researchers [3-7, 10], however, participant-centered solution design approaches are not explored with similar rigor and leave space for exploration. While many of the challenges has been mentioned in prior work, understanding the intensity of barriers faced by Women in Computing (WiC) in Bangladesh is important. For instance, Iftekhar et al. highlight peer prejudice faced by women in labs, labeling women as 'less competent' [10]. However, studies from other Global South regions, such as India and Namibia [25, 108, 109], do not report similar prejudices. The impact of such biased comments on female peers in Bangladesh remains unexplored. While previous research indicates structural exclusion from advanced training and evening lab access, the students' perspectives on the intensity of this exclusion are missing [3-7]. Given the exclusionary gendered experiences from close and respected individuals like family and faculty, an indirect method was necessary to capture these experiences. To understand the subtle social barriers women face, we framed our exploration in context by asking participants through fictional inquiry (FI) methods [19]. We wanted the participants to bypass existing socio-cultural norms or structures by prompting them with partially fictional artifacts or scenarios used in FI providing the participants a non-intrusive space to express themselves. By doing FI, we found strong emotional responses towards authoritative figures such as teachers and parents, and infrastructural issues of educational institutions. To enrich our analysis, we used in-depth analysis of the FI responses using qualitative methods as well as Natural Language Processing (NLP) based analysis to visually understand the overall scenario. NLP has significantly advanced qualitative research across various domains [91-96]. Sentiment and emotion mining, specific NLP techniques, uncover underlying intentions and opinions in data. Traditional text analysis interprets content, while sentiment analysis reveals emotional subtext. In our study, we used NLP models to analyze participants' quotes from letters, alongside a codebook annotated by qualitative researchers for detailed analysis and visualization. This approach clarified the emotional expressions and sentiments in the letters. After identifying perceived barriers through FI, we co-designed solutions with female computing students, incorporating their voices [1]. This process allowed participants to envision solutions beyond existing limitations, highlighting the priorities and challenges of Women in Computing (WiC).

We involved 48 female undergraduate participants from computer science and engineering disciplines across Dhaka, Rajshahi, Sylhet, and Chattogram divisions. Each region's participants engaged in two FI workshops,

sharing challenges through letters to different personas, followed by a co-design workshop to develop solutions. While FI-based methods and co-designing solutions have been used in education [20], they have not been applied to address WiC issues. Participants proposed various solutions, including policy and technical interventions to enhance the learning environment, increase family and societal support, and improve transportation facilities. This research incorporates the voices of female undergraduate students, presenting their ideas to the HCI community without making specific design recommendations.

While existing literature includes qualitative studies on WiC in Bangladesh, our novel use of FI employs indirection and non-verbal communication to elicit responses participants might fear to express publicly due to social prejudices. Using NLP based exploratory visualizations, we uncover hidden sentiments of fear and sadness from FI workshop letters. Our work brings together FI and NLP in a context where such interdisciplinary combinations remain rare, particularly in studies on marginalized women in computing in the Global South which opens new possibilities for investigating socially taboo topics among marginalized communities. Rather than applying NLP models blindly, we adopted a cautious, context-sensitive approach grounded in qualitative insights. Aware that most pretrained models are developed using datasets that underrepresent Global South contexts, we treated their predictions as one signal within a broader analytic process. Our workflow began with manual review of participant language to identify recurring patterns, linguistic markers, and cultural references. We then used existing sentiment and emotion classification models to analyze the data but interpreted the outputs through a comparative lens—crosschecking them against human annotations informed by domain expertise. This allowed us to identify systematic misclassifications, particularly where models struggled with sarcasm, mixed emotions, or cultural nuance. We integrated qualitative validation and domain-aware evaluation to show how NLP tools can be applied responsibly in underrepresented contexts, while exposing their limitations and potential failure modes.

As we have applied Fictional Inquiry in the context of WiC in Bangladesh, our study adds to the growing body of regionally grounded design research in HCI and ICTD. It surfaces participant-defined priorities and structural critiques and shows how tertiary students are influenced in their academics by teachers, parents, and peers, complementing existing studies by illustrating the intensity of their problems and giving voice to their ideas. The research underscores the need for continued efforts to support women in computing in South Asia, ensuring that any girl can aspire to this field without prejudice.

This paper contributes to the HCI community in following key ways:

- Firstly, we adapted Fictional Inquiry (FI) to the context of Women in Computing (WiC) in Bangladesh, where it remains underexplored. While FI has been used in education and HCI for discussing sensitive issues [20, 112], our use focuses on gendered constraints in computing education, providing a new application in this cultural and disciplinary context. We aimed to help participants to express their feelings non-verbally, under the disguise of communicating with fictional personas while writing letters to imaginary characters designed from the societal and historical elements presented in the related context.
- Secondly, this work presents a user-defined, solution-focused approach using co-design workshops to address societal, structural, and resource-related challenges faced by women in computing. Female tertiary-level students proposed tangible interventions – through policy suggestions, through drawings, app layouts, and flowcharts, to improve conditions for women in computer science. This contrasts with traditional approaches where researchers recommend solutions on behalf of participants.
- Thirdly, we used NLP models to explore the data visually, such as sentiment and emotion analysis to examine the emotional tone of the letters and workshop discussions. While general models have limitations when applied to culturally situated texts, they helped us identify broad emotional trends. These visualizations supported our qualitative coding and interpretation, allowing us to explore the emotional depth of responses expressed through both language and non-verbal forms.
- Finally, we present our methodological reflections and lessons learned, shaped by our reflexive analysis of our own positionalities as NLP and HCI researchers. These include practical insights on using NLP tools with cultural sensitivity, recognizing model biases, and exploring fictional inquiry as a

participatory method for imagining more inclusive policies. These reflections are especially relevant for feminist and cross-cultural HCI and NLP research.

2 BACKGROUND: TERTIARY EDUCATION LANDSCAPE OF BANGLADESH

In our work, we focus mainly on the context of Bangladesh, a densely populated country of 173 million people that have almost equal gender distribution [51].

2.1 Challenges in Generic Technology Access and Exposure

Socially there is practice of patrilineal descent (influence of patriarchy), and patrilocal residence (where women move to live with their husbands and relatives) [52]. Studies have shown how social norms and religious beliefs limit women's familiarity and use of technology [50]. Apart from social challenges, there are infrastructure barriers for example, in rural areas, internet connection is poor, paired with travel restrictions hampering access to publicly available internet access points like tele centers or internet cafes. In addition, sometimes there are restrictions on women on use of personal mobile phone [111]. The lack of exposure, unequal distribution of household chores, contribute to their lack of confidence in their capacity for science and technology contribute to the gender based digital divide [42]. Women are often more vulnerable to technology mediated abuse such as cyberstalking, impersonation and personal content leakage as a result of this [12, 14, 15, 64]. In addition, women are expected to behave in certain ways online and as a result many hide their identities or sentiments [9, 12, 13, 15]. Students' educational experiences are also harmed by these infrastructure problems [16, 18]. However, our work here focuses on women in computing, where they are assumed to have the capability to navigate the technology space more fluently.

2.2 Women in STEM and ICT Education:

According to Global Gender Gap Report 2023, Bangladesh ranks first in South Asia in terms of gender parity and 59th globally, however the educational attainment subindex reveals a persistent gap in enrolment in *tertiary education* [55]. The country has a high dropout rate in the transition from higher secondary to tertiary level education as around 22.72% of students drop out after higher secondary education.

At the tertiary level, there are 164 universities offering tertiary education in Bangladesh, however female representation is low as only 37.58% of total students and 29.79% of total teachers are women. There is a limitation of data as enrollment of computing specifically was not addressed in Bangladesh Education Statistics 2022 [56]. Gender-based socioeconomic and cultural barriers like child marriage, early motherhood, and violence against women are key factors in women dropping out of higher education [49, 62].

Recent work has suggested although women students' enrolment at the tertiary level has increased over the years, their representation in STEM fields remains low for multiple reasons including societal perceptions, poor infrastructure and prejudice [53, 58]. The Government of Bangladesh has introduced gender-sensitive policies to increase female enrolment which is achieved in Primary and Secondary education levels [53, 54]. The country has ICT education mandatory from grade six to twelve, but these programs have not focused on computing education in tertiary level [53, 57]. "Digital Bangladesh", an initiative to promote ICT accessibility in all areas of Bangladesh was introduced in 2008 [63], but these are generic initiatives not focused on computing students.

The first formal computer science program in Bangladesh was introduced in 1982 in BUET [59]. Currently, there are many institutes in Bangladesh that offer computer science and engineering degrees, however only few of them are accredited by Board of Accreditation for Engineering & Technical Education (BAETE) [60]. Previous work showed that enrolment in CS majors, continuation of study, involvement in competitive programming and future plans are highly impacted by the mindset, economic and sociocultural position of the belonging family [7].

3 RELATED WORK

Exploring the status of women in computing in the Asian region in comparison to Bangladesh will help us to contextualize the current problems.

3.1 Status of Women in Computing in Asia

For underrepresented regions, the literature often focuses on general ICT access and training, instead of women in tertiary education in computing. Blum et al. found that *“the notion of a gender divide in how men and women relate to computing, traditionally attributed to gender differences, is largely a result of cultural and environmental conditions. Indeed, the reasons for women entering – or not entering – the field of CS have little to do with gender and a lot to do with environment and culture as well as the perception of the field”* [25]. In many South Asian regions Women's education is often viewed as unimportant, and families under-invest in women's education [30].

Among Asian countries, there is significant diversity in the participation and engagement of women in computing depending on the country and the culture. Being attracted to the stability and the well-paid jobs in engineering and medicine, many parents push their students to enroll in computing for CS and IT jobs in India [26, 28, 29]. However, the safer IT offices are not safe since there are security concerns when women travel and for work shifts that are late at night or late because of physical safety issues [26, 27]. Parents often create pressure of marriage to female students as a solution to social problems [26].

However, not all countries are fortunate enough to have a strong representation of women in the ICT field. Research on Nepal shows that issues like a lack of gender-sensitive ICT policies, high female poverty, and the perception that ICT careers are more suited to men hurt the engagement of women [40]. A study on technology access and usage among women of Afghanistan mentioned while the women have strong familiarity with technology, they face challenges like inadequate ICT laws for handling online abuse, social restrictions on visiting public internet cafes [41]. Bangladeshi women also face many similar challenges.

Another perspective is however seen in many Muslim countries like Malaysia, Kuwait and Saudi Arabia which show high representation of women in computing and have markedly different gender perceptions around having a career in computing. Computer science is not seen as a masculine or feminine field and the perspective is gender-neutral in these cultures [31, 32, 34, 35]. Families encourage women to enroll in computing for diverse reasons, including intrinsic interest, higher income, government scholarships, and increased job opportunities [31, 32, 34, 35]. Particularly fathers often encourage their daughters to enroll in computing for financial and security reasons and indoor office job opportunities [31, 32]. A culturally positive perception of women in computing, optimistic attitude, female role models as well as government policies on the education of women with a focus on ICT and quota systems helped Malaysian women to make strides in computing [31, 32, 33]. Although some gendering was seen even in favored conditions, such as physical activities like working with electronics, networking and civil engineering majors where fieldwork is involved were looked upon as “masculine,” in contrast to software engineering and programming which are considered theoretical and fit for women [31, 32]. In some Muslim countries faculties can be both female and male, but in classroom settings, female-only gender-segregated classrooms are used [34, 36]. Our study, in contrast, involved mixed-gender classrooms at the tertiary level and a similar work environment having participation of all genders. The solutions suggested by participants considered such environment over gender-segregated classrooms in our findings.

3.2 Women in Computing in Bangladesh

Research on women in computing in Bangladesh shows challenges women face including consistent underrepresentation of women in STEM education and related jobs, lack of mentors along with the scarcity of female role models, unsupportive and exclusionary behavior from male peers, challenges related to security, transportation and accessibility related issues which become obstacle to a career in computing [3-7, 10, 16-18]. According to a survey of ICT industry in 2018, only 16% of the total ICT workforce were women [61]. Table 1 shows major work exploration from challenges and proposed solution perspectives from major works focusing on women that includes study on Women in Computing (WiC). These studies use methods such as surveys interviews, fictional enquiry, focus group discussions and various interactions to engage with researchers to identify challenges. The challenges explored consist of societal ones as well as infrastructural ones that limited WiC in various ways as presented in Table 1. Finally, the possible solution approaches cover the ranges of ideas suggested or practiced by the researchers. Our work, on the other hand, tries to focus on barriers in ways without directly communicating to WiC where they share their current challenges through letter writing and

the possible solution approaches are again suggested by the same group of participants through design workshops varying from existing work methodologically focusing on the same problem space.

Table 1. Related work comparing women in Computing in Bangladesh

Focus (with respect to WiC), Work, Date [Ref]	Challenges	Solutions	Solution Design Entity
Focusing on laboratory education 2014, Survey (N=117), FGD (N=40) [10]	Being judged, lack of competency, humiliation.	Tries female only educational activities.	Suggested by researchers.
Focusing on Women in industry 2019, interview (N=35) [7].	Poor work environment, work-family balance	Enforcement of equality from constitution	Practiced by researcher.
Focusing on students and parents in CS education, 2019. Interview parent (N=7), student (N=10). [5].	Lack of family and institutional support; infrastructure problems.	Supportive family members and institutions.	Suggested by researchers.
Focusing on learners 2020. FGD (N=12), Interview (N=7), Group discussion (N=100)/ [110].	Lack of family and institutional support; infrastructure problems, sexual harassment.	Extra sessions for practice of coding for women.	Suggested by researchers.
Focusing on students and business-women, 2022, FGD (N=30). [18].	Lack of opportunity for networking and learning.	Networking and skill enhancing initiatives.	Practiced by researcher.
Focusing on professionals, 2023. FGD (N=42). Interview (N=4). [3]	Social perception, work family balance, safety.	Safety measure, awareness generation.	Suggested by researchers.

4 METHODS

In this research work, we have used two workshops using fictional inquiry (FI) to explore the existing challenges women in computing face. This was followed by a co-design workshop to develop the possible solutions for the barriers by the same group of participants. We present the details and rationale of these techniques in the following subsections.

4.1 Fictional Inquiry and Co-Design

Fictional Inquiry to Explore Challenges: In this work, we use FI where participants are able to bypass existing social challenges as according to Dindler, “*The central idea of FI is to temporarily change or bypass existing socio-cultural structures in a given practice...FI allows designers to confront users with their own practice*” [19]. FI uses simple ‘props’ to create partially fictional scenarios and artifacts to let the participants explore the world through a fictional lens which assists in coordinating or prompting the imaginings of the participants [19, 21, 22]. It has been used in various contexts such as education setting for co-designing with preschoolers [20], reimagining museum experiences [22], designing city games with marginalized communities [21] and producing smart materials for future [22]. Existing work in education using FI showed enhanced self-expression, liberation of fear of self-exposure and stimulation of creativity [24] of students which we have used as our inspiration.

Fictional Inquiry Props: In our FI workshops, we engage the participants in a conversation to fictional characters to elicit strong emotional responses with letters as our ‘props’ since a letter allows a personal space for individuals. We used a format of letter writing where letters arrived to participants from two different personnel where participants responded individually by writing their own letters to these personas.

Fictional Characters: These fictional characters were created considering local cultural context of Bangladesh. The group of researchers worked together through their lived experiences to design the letters in a collaborative manner. One fictional character named *Oporajita* represented the support and mentorship women receive from an imaginary female supportive character, written with a soft tone seeking their challenges (details are provided below describing FI 1). The second fictional character, *M3*, was a male who questions women’s intelligence and

ability to work outside written in a harsh tone showing authoritative position presenting negative mindsets exhibiting traditional stereotypes in our society. This is designed based on challenges referred to existing literature and the first authors decade long experience of working for women in computing (details are provided below describing FI 2).

Testing of FI Letters: The letter was initially tested by one phase of response collection and analysis. The researchers along with two volunteers who were not involved in the research responded to the letters. Based on the suggestions of all the responders, the letters were modified.

Rationale: The letters written to fictional characters are expected to create a response without any existing social obligation one may feel so that the participants could share their thoughts to these characters without any fear or prejudice of hurting their feelings. The varying personalities were employed to provide both supportive and negative tones to bring out different emotional responses from participants. Although there have been regional explorations on barriers women face in the computing field, this research wanted to reach further using fictional inquiry since it allows the participants to brainstorm within the context of an imagined reality [19, 20].

Co-Design as a Solution Approach: We wanted to engage the participants actively in designing a solution that is appropriate in their socio-cultural context taking into account existing constraints using the co-design method. It is used “to refer to the creativity of designers and people not trained in design working together in the design development process” [1]. It further considers, “users as expert of their experiences” [73].

In our work, we wanted to design new solution approaches with our participants considering the challenges they explored through FI workshops. Combining these two methods, we were able to get participants to generate solutions grounded in their current environment.

4.2 Participant Recruitment

We used purposive sampling to recruit participants from four different divisions of Bangladesh - Dhaka, Rajshahi, Chattogram, and Sylhet; by approaching key informants (KI), a person known and trusted by the participants [72]. The researchers reached out to the KIs through snowball sampling [115]. The researchers, KI and the participants had an introductory session before the workshops establishing a relationship. The participant information is given in Table 1. The information is provided without referring to their specific institutes to ensure their identities are protected, considering the limited number of female students in Computer Science and Engineering programs.

We shared the purpose and moderation of the study with the KIs and asked them to find female students studying subjects related to computing at the tertiary level - in all the cases it was students studying Computer Science and Engineering. The KI explicitly stated the study purpose to the participants before the workshops. Students who showed interest after having all the information on the study participated in the workshop. Researchers consulted the KI to select the study location. The participants from every region participated in three consecutive workshops named Oporajita, M3, and Co-design of solutions as presented in Table 2. The participants were recruited for three consecutive workshops ranging from second year to final year in undergraduate studies except for Dhaka. In the first study taking place at Dhaka, it was clear that the first year students were not exposed to all the type of classes (e.g., lab and theory).

4.3 Workshops

There were three workshops – FI 1 Workshop and FI 2 Workshop to illustrate challenges through responses from two fictional characters Oporajita and M3 that has a positive tone and negative tone respectively followed by CD: Co Design Workshop 3 focusing on the solution aspect (the letters are provided in Appendix A). Each workshop was conducted on separate timeframes with the same participants, the time distribution is presented in Table 3. Workshop are explained in detail later. The FI 1, FI 2 and CD workshop took place in consecutive order starting with a positive tone, then through a negative one and finally on a separate day brainstorming on possible design approaches. The workshops all took place in separate days except for the ones in Sylhet where the researchers conducted two FI workshop in the same day and realized that it was tiring process and ensured separate days for each workshop. Each of the workshops ran once in each region with all the participants where

the individual workshop details are given in the following subsection. A pictorial presentation of the workshops is presented in Appendix B.

Table 2. Participant Information

Region	N	Gender, year in Computer Science and Engineering (CSE) Program	# of Workshops. Frequency of taking place. Challenges	Workshop details FI: Fictional Inquiry; CD: Co-design (time hours)
Dhaka	11	Female, 1 st to 4 th Year students *	3. Took place in 3 consecutive days. Not experienced.	FI 1: Oporajita (2-3) Day 1 FI 2: M3(2-3) Day 2 CD (3- 4) Day 3
Rajshahi	10	Female, 2 nd to 4 th year students.	3. Two took place in a single day: (in between 2 hours break) and one in the following day. Not experienced.	FI 1: Oporajita (2-3) Day 1 (morning) [Break (2 hours)] FI 2: M3(2-3) Day 1 (afternoon) CD (3- 4) Day 2 (morning)
Sylhet	16	Female, 2 nd to 4 th year students.	3. Took place in 3 consecutive days. Finding a meeting place.	FI 1: Oporajita (2-3) Day 1 FI 2: M3(2-3) Day 2 CD (3- 4) Day 3
Chattogram	11	Female, 2 nd to 4 th year students.	3. Took place in 3 consecutive days. Finding students.	FI 1: Oporajita (2-3) Day 1 FI 2: M3(2-3) Day 2 CD (3- 4) Day 3

Table 3. Workshop Information

Study Type	Description	Max Time
FI 1: Respond to a letter from Oporajita, positive tone	Receiving individual letter from Aparajita	60-90 Minute
	Participants responds to individual letter	
	Group Discussion on challenges participants faced and shared in letters	25-30 Minute
FI 2 : Respond to a letter from M3, negative tone	Receiving individual letter from M3	60-90 Minute
	Participants responds to individual letter	
	Group Discussion on challenges participants faced and shared in letters	25-30 Minute
CD Workshop: Co design solutions.	Group discussion on major challenges identified from the letter responses in FI workshops	20-25 Minute
	Design concept and exploration by individuals working together in the same space.	90 Minute
	Group Discussion with conclusion on solution approaches	60-90 Minute

FI 1: Responding to a Letter from Oporajita, a fictional female character: This letter was from an imaginary person named Oporajita, a female descendant of Begum Rokeya. Begum Rokeya is a prominent writer, educator and thinker who contributed towards the empowerment of Bengali Muslim women and known as a significant female role model for Bengali women [77].

Oporajita, the female persona shared a bit of Begum Rokeya's struggle in the past and told the participants she wishes to continue Rokeya's will to empower women in Bangladesh. She candidly asked about how women today are doing as she is hopeful that women have progressed a lot over all these years. We expected the candidates to hold knowledge about Rokeya as even in our primary education curriculum her stories are included and framed Oporajita as a descendant of her lineage to establish a connection of trust and respect. The letter was written in a supportive tone. The goal of this letter is to explore the current barriers female students are facing where they can respond to an imaginary supportive role model who envisions women's freedom through knowledge acquisition. The letter is attached in the Appendix section. The workshop participants were asked to read the letter and responded to the letter individually in a written format without any given word limit or page limit. It was followed by a sharing session where all participants from the same region joined. The discussion was moderated by a researcher to ensure the presence of all the voices. The workshop provided

necessary stationaries and supplies for the participants to respond and brainstorm including papers, pens, pencils, whiteboards, poster boards, color pencils, markers etc.

Workshop 2: Responding to a Letter to Mustakim Muhammad Monsur (M3), a fictional male character: This letter was from a male imaginary person M3 who holds a position of influence to make changes for working women. However, in his letter, he is skeptical about women's ability and desire to join the workforce. He refers to many of his own female family members who are more interested in other things rather than pursuing a career. The letter is written in a harsh tone with judgment. The goal of this letter is to understand the challenges women face after negative emotional feedback is given. As discussed in section 2.2.2, in existing literature it was found that women in Bangladesh face discriminatory behavior from male authority figures and peers in both education and work life. We asked the participants to respond in their letters without any fear, as in real life they may not have the courage to argue against such people. Similar to FI1, there was no word limit to respond to this letter. The letter to M3 is attached in the appendix section. The supplies were similar to ones provided in FI 1.

CD: Co-Design of Solutions: The co-design workshop oriented around the problems discussed in the previous workshops. Pre-workshop analysis was done on the responses of the participants in Workshop 1 and Workshop 2 to identify the key challenges mentioned. These challenges served as a starting point as we presented them in a presentation slide to the participants. Then the participants focused on designing ways to address solutions given as an open ended problem through co-design, a technique used as a response in other regions such as Namibia [109]. We allowed space and time to think about solution approaches towards challenges and provided the participants with a floor to have an open discussion among themselves. Many participants preferred technology solutions, being technology students. It was a full day workshop. This workshop was conducted in spaces agreed upon with the participants that had working space for design and brainstorming. The design session used the co-design approach where the participants engaged with researchers to work on solutions. It reflects on an active solution approach since there is a gap in having engagements coming from participants being women in computing.

Since the participants were undergraduate students familiar with general software design and visualization techniques taught to them in prior courses, many of them preferred creating solutions around their expertise - designing algorithms, flowcharts, visualizations and diagrams with stick figures during the sessions. Some of the participants initiated their design from their imagination of ways existing barriers can be eradicated where some worked on policy level changes while others outlined a technological solution. However, most of the participants preferred to present their solution as lists.

4.4 Qualitative Content Analysis

We have conducted a few rounds of content analysis - initiating with transcribing the audio recorded during the workshops and coding the initial contents consisting of letters and discussions. The audio recordings of the discussions during and after FI workshops and the discussions during the co-design workshops (accumulating to 5.45 hours of audio) were transcribed. After the initial coding process, we categorized the codes and refined them through several iterations. To identify the key themes, we used reflexive thematic analysis process [76]. The qualitative researchers engaged in conducting the workshops worked collaboratively towards data analysis.

The analysis evolved into major initial codes as academic, teacher, advertisement, awareness, bias, exposure, family, forum, infrastructure, mentorship, networking, security, special situation, transportation, security, and underrepresentation (presented in Figure 1-Right). We present the regional frequency and overall frequency of the codes and overall frequency here in Table 4. The content from the problem side came mostly from the letters written to Oporajita and M3 in FI 1 and FI 2 the Fictional Inquiry workshops respectively. The solution related discussion mainly came from the CD design workshop, some of it was also suggested in the letters of FI workshops.

Table 4. Codes and Frequencies

Codes	Frequency (Challenge domain)	Frequency (Solution domain)	Frequency (Total)
Academic	18	13	31
Bias	23	6	32
Family	22	7	29
Forum	N/A	8	8
Infrastructure	14	7	21
Mentorship	N/A	11	11
Networking	N/A	8	8
Security	11	5	16
Transportation	11	8	19
Under representation	15	N/A	15
Awareness	9	27	36
Achievement	10	N/A	10
Advertisement	N/A	10	10

For individual themes, we contrasted the challenges along with the solutions and cross aligned them with the participants' information, their regions and how the concept emerged through a letter to Oporajita, to M3, or during the design session. We created individual persona (as presented in Figure 1-Left) regional persona (as presented in Figure 1-Middle) to reflect findings from challenges and solutions space from that particular region. We further analyzed the data to show the time references of the topic whether it refers to levels of primary schooling, middle school, high school, tertiary level, professional level etc. Through the analysis process, the participants' concepts were considered through various aspects such as individually; according to their study locations; based on their original residences and based on other personal interests to thrive etc. We also took into account the observations and personal notes researchers took during the study process.

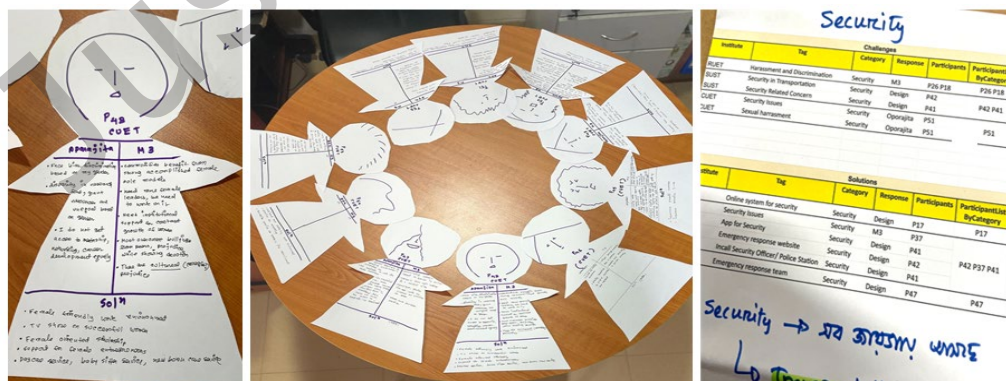


Figure 1. (Left) Individual participant content analysis (Middle) Region-wise content analysis of participants (Right) Theme-wise analysis based on challenges and solutions

For this particular study, we focused mostly on the solutions suggested by the participants and worked through the solution space suggested by participants. Then multiple themes were converted into three broader themes

covering *academic, social and security-related aspects*. Security was a separate topic as it emerged in both academic and social context.

4.5 Natural Language Processing Based Analysis

Traditionally, qualitative research involves manually coding interview transcripts, user observation notes, and survey responses, however natural Language Processing (NLP) can be a powerful tool to enrich qualitative research by offering new ways to analyze and understand user data [91-96]. NLP techniques like sentiment analysis, emotion mining, and named entity recognition can augment the qualitative research. While NLP models are not perfect, and their accuracy can vary depending on the quality and size of the training data and researchers need to be aware of potential biases and errors in the NLP outputs and conduct human review to ensure the validity of the findings, we can identify recurring themes, participant frustrations, and positive or negative experiences within the data. Previous work suggests mixed methods research, drawing from both NLP and HCI can allow us to develop richer and holistic understanding of the narratives given by participants in qualitative research to analyze larger patterns across stories [78].

Transfer learning is an approach in natural language processing which saves time and resources by starting with a pre-trained model that already has learned valuable features about language. [ref] Training a sentiment analysis model from scratch requires a large amount of labeled data which can be expensive and time-consuming to collect and annotate. Transfer learning allows us to leverage pre-trained models on massive datasets (which are often available only in English) that capture general language understanding. which can then be adapted to the specific task. In prior work sentiment analysis has been approached with finetuning pretrained models for specific tasks in both Bangla, English/Multilingual context [79,80].

Our dataset constructed from interview transcripts is small, so instead of training models from scratch we leveraged English and multilingual pretrained models finetuned on sentiment analysis and emotion mining in our tasks to get the relevant predictions. Our pipeline of sentiment and emotion analysis involves collecting participant responses via conducting fictional inquiry and co-design workshops, then annotating the quotes from the letters by grouping them into codes and themes, preprocessing the data by cleaning and translation, choosing relevant pretrained models for our analysis, getting the sentiment and emotion scores and analysis of the outputs via data visualization techniques. Relevant models along with their data labelling schemas are described in the following sections.

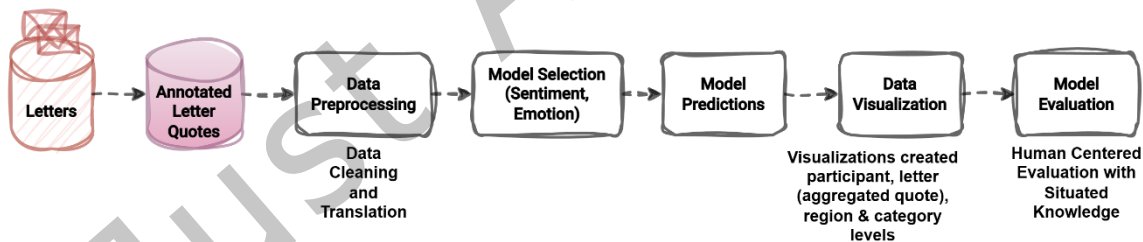


Figure 2. Pipeline of text mining of fictional inquiry letters

Annotated Letter Dataset Construction from Participant Responses: To prepare our dataset for analysis with natural language processing techniques such as sentiment analysis and emotion mining, after performing qualitative coding and assigning relevant tags to the interview transcript quotations, we prepared a dataset from our interview transcripts for NLP work. This dataset included following features - participant id, participant location, participant quotes from letters sent to M3 and Oporajita in workshop 1 and 2, quotes from design sessions in workshop 3, primary category/code assigned to the quotes along with sub-categories of challenges and solutions.

Relevant metrics such as number of quotes and tags for this dataset is given below in the following table X. The tags assigned to each quote is the most granular unit of our coding scheme, these were further grouped into relevant sub-categories and primary categories. Our aim was to clearly differentiate whether the sentiment and the emotion changes when participants express themselves via letters to M3 and Oporajita, two unique

individual personas we created for fictional inquiry workshop. We also additionally consider the design workshop related quotations for comparison.

Table 5. Letter dataset metrics

Dataset Features	Metrics
Number of Quotes of M3 letters	148
Number of Quotes of Oporajita letters	136
Number of Design Quotes	195
Number of Tags of M3 letters	68
Number of Tags of Oporajita letters	78
Number of Tags of design solutions	144

Data Preprocessing: Since the letters were written in both English and Bengali, we used minor preprocessing before using sentiment analysis and emotion analysis models. We used language detection tools to detect Bengali along with translation models for translating letter quotes from Bengali to English. A summary of the tools and models used are given in Table 5 and Table 6 respectively.

Table 6. Tools used in data processing

Tool/Model	Purpose
Langdetect [97]	Language Detection
PyTorch [98], HuggingFace Transformers [99]	Modelling
Googletrans [100], BanglaT5 English to Bengali Translation Model (csebuetnlp/banglat5_nmt_bn_en) [101]	Bengali to English Translation
Twitter-roBERTa-base for Sentiment Analysis [81]	Sentiment Analysis Model
Emotion English DistilRoBERTa-base [84]	Emotion Analysis Model
Pandas [102]	Data Processing and Analysis
Matplotlib [103], Seaborn [104]	Data Visualization

We selected open-source tools from the HuggingFace repository based on accessibility, free in price, reported performance, and transparency of pretraining data. Here we document the models selected for sentiment and emotion analysis.

Sentiment Analysis Model Selection: Our experiments used the “Twitter-roBERTa-base” model [81], a RoBERTa-base model trained on approximately 124 million tweets from January 2018 to December 2021 and fine-tuned for sentiment analysis on the TweetEval benchmark [82,83]. The model incorporates preprocessing steps to reduce potential spam and near duplicate tweets and employs a continual training procedure. Since the model was originally trained on Twitter conversational data, it is expected to be relatively suitable for analyzing our dataset, which also consists of conversational content, although some domain differences may persist. Sentiment labels produced by this model are negative (0), neutral (1), and positive (2). The model operates on English text. Its validation accuracy is reported at 71% with an F1 score of 71% [91].

Emotion Analysis Model Selection: For emotion analysis, we used the “j-hartmann Emotion English DistilRoBERTa-base” model [84], which scores participant quotes according to Ekman’s six basic emotions: anger, disgust, fear, joy, sadness, surprise and a neutral category [85]. This model was trained on around 20,000 samples drawn from multiple emotion detection datasets, including CrowdFlower [86], MELD [87], and GoEmotions [88]. The reported accuracy of the model is 66%, which is substantially higher than the random baseline of 14% (1/7). The model is also trained exclusively on English data. Prior studies have applied this model for rumor-related reaction analysis [89] and emotion classification of newspaper headlines [90].

Context-Aware Sentiment and Emotion Analysis: While we aimed to explore the dataset through a natural language processing lens, we remained conscious of the limitations of existing NLP models, particularly the potential biases embedded in their pretraining datasets due to demographic mismatch and model assumptions. Thus, prior to applying computational models, we conducted qualitative analysis of letter quotes to identify emerging trends and patterns in participants' expressions. We then applied pre-trained sentiment analysis and emotion classification models to generate prediction scores for each letter unit (quotes). The computational analysis was structured across multiple levels of granularity: individual participant responses, regions (location), thematic categories and sub-categories (specifically examining challenges faced by women in computing and their proposed solutions), and per-letter sentiment and emotional trajectories. The multi-level visualization approach allowed us to capture both individual experiences and collective patterns in participants' affective responses to the gendered personas. This allowed us to explore whether sentiment or emotional tones varied systematically by theme or participant, and how model predictions aligned, or misaligned, with the nuances identified in our manual analysis.

Human-Centered Model Evaluation with Situated Knowledge: We approached model evaluation through a reflexive, context-aware lens that acknowledged our positionalities as NLP and HCI researchers working across cultural and linguistic boundaries. To ground our analysis in lived experiences of participants, we manually annotated 20% of the letters using annotators fluent in local computing discourse, colloquial language, and the gendered sociopolitical context and compared model output against these human judgments. Our evaluation criteria specifically examined errors introduced during translation from Bangla to English, instances where model predictions misaligned with human annotations (including misunderstanding participant intent, flattening emotional tones, and missing cultural context), and systematic patterns of misrepresentation. The annotators were computer science students themselves from different backgrounds helped to surface errors that standard benchmarks often miss, including misclassifications of sarcasm, erasure of emotional intensity, and cultural mismatches in affect lexicons and systematic patterns of misrepresentation. We documented them through detailed case studies with direct quotes and justifications from annotators.



Figure 3. Pipeline of Sentiment and Emotion Analysis Model Evaluation

To investigate the source of errors, we also performed a pretraining data audit of the models used. Since NLP systems are trained on human language, they inevitably inherit social biases, such as those related to gender or ethnicity, embedded in the pretraining data, which is again a snapshot of the time (when the data was collected), place (source of data) and assumptions held by the NLP researchers themselves, as language is inherently situated and shaped by specific cultural and historical contexts [143,144]. This perspective informed our use of the Situated Data, Situated Systems framework [134], through which we used its guiding questions to critically examine how inherited biases surfaced in our dataset.

4.6 Research Ethics and Incentive

The study is approved by the Institutional Review Board (IRB), (details redacted for blind review). On the day of the workshop, we verbally explained the workshop objective, process and permission required for recording data and photographs. The participants were adults and verbally agreed and signed a written consent form. All notes, recordings, photographs and writings are stored in a secured drive which can be accessed by the researchers only.

4.7 Research Team and Positionality

All the researchers in the team were born and brought up in Bangladesh and are familiar with barriers women face being part of Computer Science and Engineer community. The researchers are sharing their background and experiences based on Ruth Franeker's views for readers to understand the importance of reflexivity in qualitative research [75, 116]. All of the team members are qualitative researchers with more than 3+ years of research experience and have undergraduate degrees in computer science from Bangladesh. They have worked on diverse projects spanning machine learning, human computer interaction research focusing on marginalized communities as well as low resource natural language processing. On the other hand, the quantitative research team for NLP did not have any direct interaction with participants and worked on the collected dataset from qualitative researchers. All the researchers are trained with human subject handling as required by the IRB process. The team composition was balanced in terms of gender, religion and experience of growing up in urban vs rural areas. The team is familiar with the contextual environment of the student participants because they graduated from local universities and went through similar curriculums, thus they have lived experience of existing challenges and solutions.

To avoid the social desirability bias [114], the researchers allowed *anonymity* and complete *confidentiality* of the participants in the FI workshops. We used an *indirect* inquiry method FI and prepared two different personas for participants to express their viewpoints via letters. It was further ensured that there was no direct interview process in the study. The research took place mostly through active participation of tertiary level students through their writing and design process which was discussed later with the researchers after an initial introduction. The lack of direct interaction between researcher and participant during the exploration process minimized the possibility of social desirability bias. The researchers were friendly and supportive of their ideas while maintaining neutrality. Faculties were not present during the interviews to avoid the possibility of the power imbalances with the participants, while the facilitators themselves were around similar age range (around 3-5 year differences).

5 PARTICIPANT PERCEPTION OF CHALLENGES AND RESPONSES

There were three stages of exploration where participants initially responded by writing letters to Oporajita who had a supportive tone, followed by another letter to M3 that was negative and harsh about women's ability and finally a design workshop that preceded the topics that came from the letters across participants. In this section we analyze the perception and attitudes of our respondents towards the fictional characters presented to them as props from both qualitative and quantitative perspective. First, we present the qualitative research observations of our respondents towards two contrasting personas - M3 and Oporajita. Then we present the results of our sentiment analysis and emotion mining. The design workshops focused on solutions to existing problems. The participants discussed the aspects, wrote notes and diagrams of possible solution approaches which are presented in the light of problems in the following subsections.

5.1 Qualitative Observation of Fictional Inquiry Workshop

The letters had varied levels of emotional engagement of participants even when the participants were aware of the letters coming from fictional characters being in a respectful interview and letter writing workshop settings. From our qualitative observations, we found that participants responded with highly intense emotions for the M3 letters. In the M3 letter we tried to aggravate the participants by using language such as women are not as "sharp" as men, lacks "natural talent" and not "serious" about computing. Some people found the negative attitude shown by M3 as personal insults and they used slang, or sarcastic language against the M3 persona to respond. M3 persona was considered as a "real person" by many students where they are clearly calling him by his name to respond to the letters with sarcastic and harsh reprimands. Some of the participants used negative words to respond to M3 about this problem calling him "narrow minded", "low" in their responses even after knowing it was a fictional character. For example, one of the participants wrote in her letter the following lines responding to M3:

"First of all, Mr. Mustakim can't think women are not better than men. He clearly thinks that women just think about their beauty, shopping bla bla bla, I hope, Mr. Mustakim thinks more carefully and

respectfully about women. Otherwise, it's his loss...he should not judge all women with his 3rd class point of views and ideas - P45".

In contrast to that, Oporajita persona was considered as a supportive female figure by almost all the participants and the students opened their hearts. Oporajita asked open ended questions such as whether the students are still facing “challenges” and “injustices” in the new era. A particular participant wrote in her letter to Oporajita:

“As a female student, one needs to work a bit harder than the male counterpart to prove their worth and that’s a bit disheartening in the era of establishing equal rights. - P36”.

The participants described numerous challenges including social injustices from parents and teachers, infrastructure related issues and workplace discrimination. Most of the challenges came out through the letters where workplace, balancing work and life and career related concerns were discussed more in letters to M3 while academic, family and social concerns came from letters written to Oporajita and M3. We wanted to expand our qualitative observations by augmenting with quantitative analysis by using the latest natural language processing models and the results are presented below.

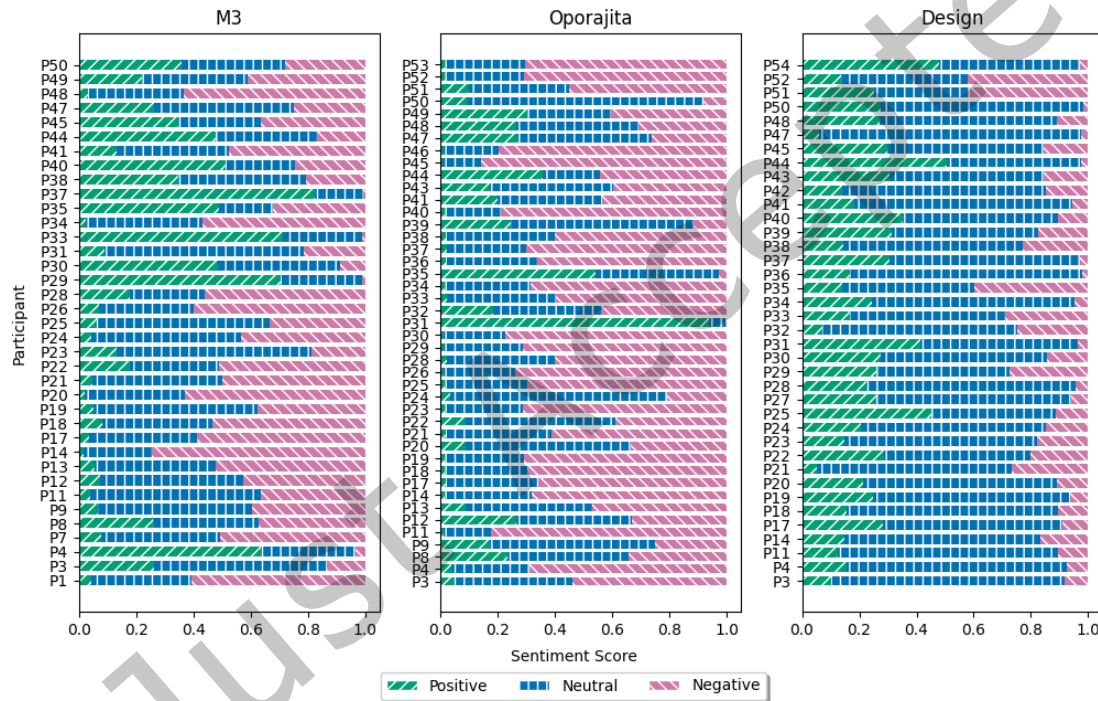


Figure 4. Sentiment Score by Per Participant

5.2 More Negative Sentiments shared to Personas

Our in-depth analysis using various tools showed how the range of sentiments varied across different letters, and also across individual participants. As can be seen in Figure 4, the expression of the letters to Oporajita and M3 persona had more negative and positive sentiments compared to neutral emotions compared to the co-design sessions, which was dominated by neutral emotions. Across responses to Oporajita and M3 personas, the number of negative sentiments expressed was higher in Oporajita responses. This suggests that the participants were more forthcoming in expressing their complaints to Oporajita, leading to a higher number of negative sentiments as compared to M3. Conversely, the participants were more focused on demonstrating their capabilities to M3, resulting in a higher number of positive sentiments. The design dataset displayed a more balanced distribution of sentiments, with most of the opinions being neutral.

We find similar trends in the regional representations of sentiments as presented in Figure 5 (a). Notably, Oporajita has a higher number of negative sentiments. Additionally, the sentiment trends for specific locations remain consistent across datasets. Furthermore, it is worth mentioning that Rajshahi exhibits the lowest positive sentiment across all datasets, while Dhaka has the second lowest positive sentiment. Conversely, Chattogram displays the highest positive sentiment across all datasets.

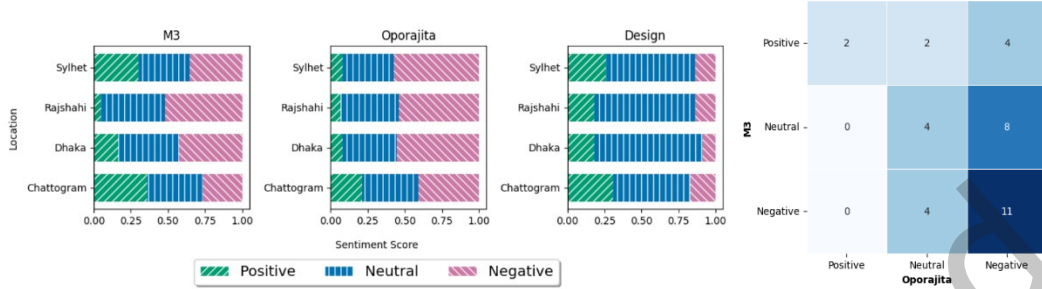


Figure 5. (a) Sentiment Score by Location (b) Confusion Matrix of M3 VS Oporajita

According to Figure 5 (b), it is evident that many participants who showed positive or neutral sentiment for M3, shows negative sentiment when it comes to Oporajita. This shows that using different personas in workshops such as fictional inquiry can result in widely different attitudes expressed towards different personas even when it comes to general sentiment scores. To create the visualization in Figure 5 (b) initially sentiment scores of each quote were identified via the sentiment scoring model. Then the quotes were grouped by the letters the originally the quotes were from, and we assigned each letter a sentiment of positive, negative and neutral based on the most frequent sentiment. Then the sentiments were compared for participants who sent letters to both M3 and Oporajita via confusion matrix.

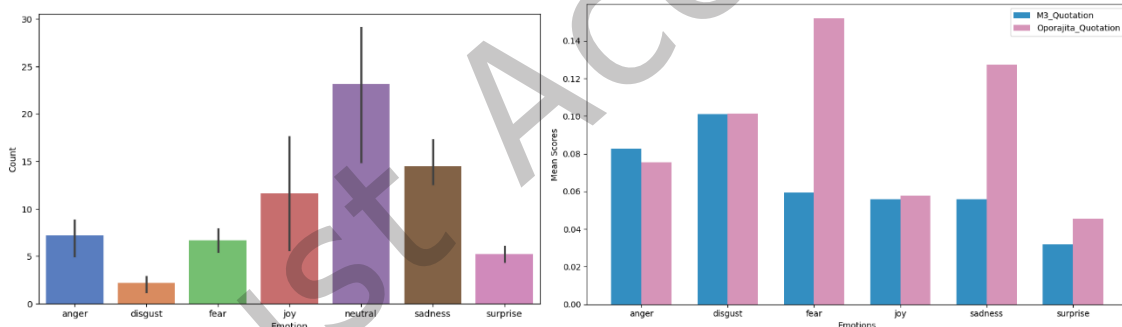


Figure 6. (a) Most Frequent Emotion of Participants for combined Quotation Dataset (b) Comparison of average emotion score for M3 vs Oporajita

5.3 Diverse Emotions Expressed Towards Personas

Like our sentiment scores, the emotion scores also differ for the letters to M3 vs Oporajita. However, with the addition of neutral class and fine-grained emotion labels, we can see a more in-depth picture of emotion from the emotion analysis. As we can see in Figure 6, while neutral is the most predominant emotion, sadness, joy, anger and fear are also expressed by many participants in the letters. Comparing the average emotion scores for M3 and Oporajita quotations, we find that on average M3 does get more anger response from participants, while expression of fear and sadness is higher for Oporajita.

In the heatmap of average emotion scores of M3 and Oporajita grouped by locations in Appendix C, we can see that distribution of emotions are consistent across all regions and aside from neutrality, sadness was the second most dominant emotion in all regions. In Figure 7 (a) and (b) we scored each participant according to the emotion most frequent in their quotes for both M3 and Oporajita. While the most dominant emotion for each participant happens to be neutral, we find a higher variety of emotions in Oporajita compared to M3. We

also found that fear was shown to Oporajita more instead of M3, possibly due to feeling of safety a supportive persona as Oporajita provided.

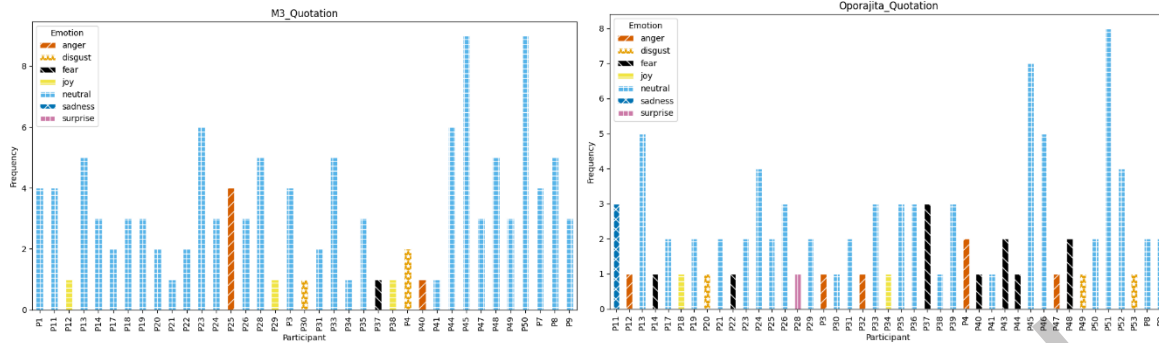


Figure 7. (a) Most Frequent Emotion of Participants for M3 quotations (b) Most Frequent Emotion of Participants for Oporajita quotations

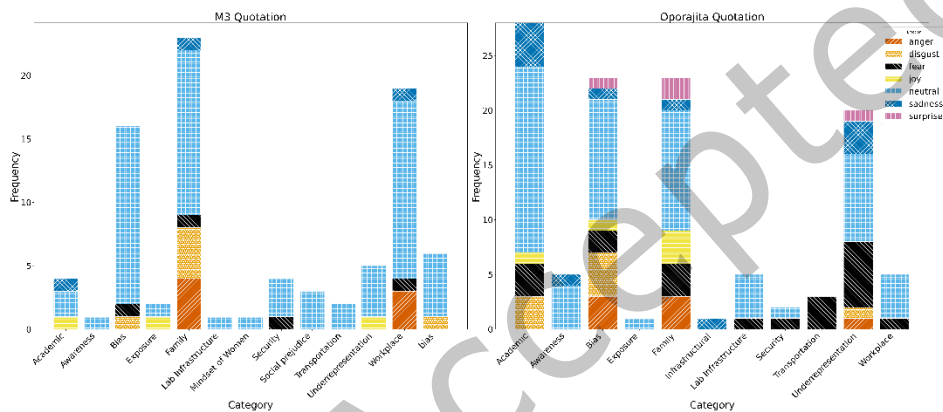


Figure 8. (a) Most Frequent Emotion of Participants for M3 quotations (b) Most Frequent Emotion of Participants for Oporajita quotations in Challenge related sub-categories

Appendix D represents frequency of Challenge related codes for Oporajita and M3 dataset across the regions for sub-categories such as academic, family problem, underrepresentation, lab infrastructure etc. When we investigate the emotions associated with each challenge sub-category in M3 vs Oporajita quotations, we can see that academic challenges came up more in Oporajita quotes along with disgust and fear related emotions. Bias category shows disgust and fear for both Oporajita and M3, but students show anger only to Oporajita. Students talked about their transportation related fears to Oporajita more too. It suggests that participants were able to talk about their fears in the supportive safe space Oporajita persona created.

While we have qualitatively coded the responses of our participants from all workshops, using NLP methods we can see that the broader trend in the dataset also follows the observations made by our qualitative researchers. Qualitative observations showed that students found the female supportive figure as a safe space while showing strong disagreement with the male persona of Mustakim. In our NLP analysis too, the students showed more diverse emotional expressions such as fear and sadness and shares their negative experiences to the supportive female figure of Oporajita, while showing anger towards M3, the dominant male figure. In the design workshops the predominant sentiment has been neutral showing the focused emotional state of the participants while constructing the solutions. Emotion analysis of the categories of the challenges also suggests participants holds a variety of emotions towards the challenges they faced while the analysis across locations shows the trends in the differences of emotions towards M3 and Oporajita are consistent across all locations. These strong emotions felt during the fictional inquiry workshops creates the motivation and the foundation to resolve the challenges in the co-design workshops where the students generated a variety of solutions while describing the numerous problems they face in their lives as women in computing.

5.4 Reflection on Power Embedded in Technologies

Justification of Human Evaluation: While we designed our interpersonal interactions to minimize hierarchical and cultural power imbalances, we recognize that power is equally embedded in the tools, languages, and infrastructures we used—particularly those related to NLP [134, 143]. Models we employed for text classification rely on pretraining datasets that reflect dominant global ideologies and linguistic norms, often marginalizing languages like Bangla or underrepresenting non-Western social contexts [142]. Translation tools, too, are rarely neutral; they prioritize standardized forms over colloquial, gendered, or context-specific expressions, potentially flattening the nuances in participants’ voices [146]. Even when models operate without human intervention, they reproduce systems of exclusion—e.g., by failing to capture the implicit meanings in locally situated speech, or by prioritizing dataset-defined “accuracy” over cultural relevance [144, 145]. This necessitates ongoing reflection not only on how we interact with participants, but also on the invisible infrastructures that mediate their data. These mismatches risk misinterpreting or erasing the lived experiences of Bangladeshi women in computing. Our team’s own positionalities—as Bangladeshi researchers with varying degrees of proximity to the tech industry, language fluency, and feminist ideology—enabled us to recognize many of these misalignments.

5.4.1 Human Evaluation of Model Responses

We have manually annotated 20% of the letter dataset (both for sentiment and emotions). After annotation we found that the sentiment model showed an overall accuracy of 73.26% (positive - 92.86%, neutral - 23.08%, negative - 95.65%) and the emotion model showed an overall accuracy of 21% for identifying emotions into 7 labels (6 emotions, 1 neutral class). Out of our full dataset, only 8 of the sentences were translated from Bangla. Our review process included annotators who were familiar with the local context (researchers who have studied/are studying computer science themselves in their final year of university or who have already graduated) and who could reflect on local context, sentiment and slang. This aligns with Buolamwini and Geburu’s call for algorithmic justice—emphasizing lived experience as a site of evaluation.

We have documented specific instances where NLP models misrepresented sentiment, tone, or cultural meaning, accompanied by direct quotes from the data and a clear rationale for why the output did not align with participants’ intent. These examples will be showcased as case studies within the paper, followed by the participatory review process that allowed annotators and community members to correct or contextualize the tool’s output. Our evaluation criteria’s included errors introduced during translation, mismatches between model prediction and human annotation (misunderstanding participant intent, flattening emotional tones, missing cultural context etc). A detailed evaluation guideline is given in Appendix E.

Translation Inconsistency:

P45(Oporajita): যদিও cse subject টা আমি পছন্দ করেই নিয়েছি, তবুও কিছু কিছু টপিকস আমার বুঝতে একটু সমস্যা হয়ই□ হ্যাঁ, আমি অবশ্যই কিছু challenges পেয়েছি, যেমন দেখা যায় ক্লাস এই অনেক কিছু অনেকেই খুব তাড়াতাড়ি বুঝে ফেলছে বা catch করতে পারছে, সে ক্ষেত্রে দেখা যায় যে আমি চাইলেও ব্যাপারটা ঠিক করে বুঝতে পারছি না□ যদিও ইন্টারনেট থেকে ঘাটাঘাটি করে আমি ব্যাপারগুলো জানতে পারছি, তারপরেও কিছু কিছু ফাঁক তো থেকেই যায়□

Translated: Although I have chosen the CSE subject, I still have some challenges to understand. Yes, I’ve got some challenges, such as that many people are quick to understand or catch these subjects in class, and that I don’t really understand them.

Review: “যদিও ইন্টারনেট থেকে ঘাটাঘাটি করে আমি ব্যাপারগুলো জানতে পারছি, তারপরেও কিছু কিছু ফাঁক তো থেকেই যায়□” - this part has been removed while translating

Model Generated Emotion Label: Neutral

Human Annotation: Sadness

Model Generated Sentiment Label: Neutral

Human Annotation: Negative

Annotation Justification:

The overall tone of the statement reflects discouragement and emotional struggle, aligning with the emotion sadness and sentiment neutral. Multiple phrases support this labeling:

- Despite the slight positive intro (“পছন্দ করেই নিয়েছি”), the core message revolves around difficulties – making negative sentiment dominant.
- “তবুও কিছু কিছু টপিকস আমার বুঝতে একটু সমস্যা হয়ই” indicates cognitive struggle and self-doubt.
- “অনেকেই খুব তাড়াতাড়ি বুঝে ফেলছে” highlights social comparison, often linked to low self-worth.
- “আমি চাইলেও ব্যাপারটা ঠিক করে বুঝতে পারছি না” suggests a disheartening gap between effort and outcome.
- “ফাঁক তো থেকেই যায়” captures a lingering sense of incompleteness, disappointment and emotional burden, reinforcing the emotional tone of sadness.

While translating to English, the translated version sounds like more matter-of-fact or neutral in tone, especially:

1. "I still have some challenges to understand" and
2. "I don't really understand them"

These phrases describe the same situation but do so with less emotional intensity – making the statement more ambiguous between neutral and sadness, depending on delivery. The tone is more objective and descriptive, using general terms like "challenges" without strongly emotional words.

Emotion Mismatch and Inability to Detect Intent:

P45(M3): And he is talking about woman's barrier! That's funny. He'll never know what a woman faces until he is a woman himself.

Model Generated Emotion Label: surprise

Human Annotation: disgust

Model Generated Sentiment Label: negative

Human Annotation: Negative

Annotation Justification:

The speaker uses sarcasm and mockery to challenge the legitimacy of a man speaking on women's lived barriers. The phrase “*That's funny*” is not expressing literal amusement but disdain – a classic indicator

of disgust. Then again, the declarative “*He’ll never know...*” reinforces a sense of frustration and boundary-setting, possibly carrying undertones of anger. This marks the speaker’s emotional rejection of the perceived ignorance or presumption, which is morally and socially rooted.

Model Misclassification Analysis:

The model likely latched onto the phrase “*That’s funny*” along with the exclamatory sign (!) as a literal expression of amusement or unexpectedness, rather than as a mockery. Without understanding tone and context, it falsely inferred an element of *surprise*.

Emotion Misalignment and Failure to Capture Underlying Tone:

P49(M3): First of all, Mr. Mustakim can't think women are not better than men. He clearly thinks that women just think about their beauty, shopping bla bla bla.

Model Generated Emotion Label: neutral

Human Annotation: disgust

Model Generated Sentiment Label: negative

Human Annotation: negative

Annotation Justification:

This statement expresses clear moral disapproval toward an individual’s sexist mindset. The phrase “*He clearly thinks...*” introduces a judgmental tone, and the use of “*bla bla bla*” dismisses the subject’s views with mockery and contempt, signaling a disgust reaction rather than mere disagreement. The speaker is not just disagreeing — they are repulsed by the idea being expressed.

The sentiment is negative, as the speaker is confronting derogatory gender stereotypes and calling out discrimination. The language conveys frustration and indignation toward an unfair portrayal of women.

Model Misclassification Analysis:

The model may struggle with sarcasm, dismissive tone (“*bla bla bla*”), or informal phrasing, often lacking training on nuanced moral emotions like disgust when they appear in conversational or sarcastic language. It may misinterpret this as neutral due to the lack of overtly emotional words, though the intent and tone are clearly emotionally charged.

Disagreement Between Annotators:

P45(M3): he should not judge all women with his 3rd class point of views and ideas. Women are empowered, he needs to learn how to respect women.

Model Generated Emotion Label: neutral

Human Annotation: disgust; **Initial Annotation:** neutral

Model Generated Sentiment Label: negative

Human Annotation: Negative; **Initial Annotation:** neutral

Annotation Justification:

This statement carries strong moral disapproval and criticism towards a sexist mindset. The phrase “*3rd class point of views*” is explicitly derogatory, expressing a sense of contempt or revulsion. Moreover, the imperative “*he needs to learn how to respect women*” reflects a moral correction, reinforcing the emotion as not just frustration but judgment grounded in values and justice. This is a hallmark of disgust over moral violation.

Model Misclassification Analysis:

Sarcasm and judgmental tone are challenging for models to detect, especially if trained primarily on formal or explicit emotional cues. The model may treat the statement as fact-stating or assertive due to its declarative structure, ignoring the loaded language like “3rd class.” Additionally, emotion models often struggle to differentiate anger from disgust, and even more so to detect disgust when it lacks overt expressions like “hate” or “gross.” This example contributes to the argument that emotion models underperform on morally charged, sarcastic, or contemptuous discourse, particularly when aimed at social injustice or gender discrimination.

Emotion and Sentiment Misalignment; Failure in Detecting Rhetoric Tone:

P17(M3): As you say “Women’s are not as sharp as men. For computing, men have natural talent.” Do you know the person who invented computer programming is a woman. She is Lovelace. Tell me if she does it with her looks. I don’t think so.

Model Generated Emotion Label: neutral

Human Annotation: disgust/anger

Model Generated Sentiment Label: neutral

Human Annotation: negative

Annotation Justification:

This statement expresses a critical response to gender stereotypes about women’s abilities in computing. The speaker is challenging a sexist viewpoint. The speaker questions whether Lovelace’s achievements in programming were attributed to her looks, implicitly critiquing the misconception that women have less intellect or capabilities than men.

1. Disgust: The speaker appears disgusted by the stereotypical notion that women are less suited for computing and are valued primarily for superficial traits.
2. Anger: The rhetorical question, “does she do it with her looks?”, expresses frustration and anger toward these reductive gender biases.

Model Misclassification Analysis:

The model's prediction of neutral emotion and sentiment indicates that it missed the underlying frustration and disgust in the statement. The model may have focused on the question structure (“Does she do it with her looks?”) and the somewhat neutral tone of the statement without fully grasping the emotional intensity related to challenging gender bias. The model might have viewed the statement as simply factual or neutral due to its question-based structure, but context (a critique of misogynistic thinking) was not sufficiently captured.

5.4.2 Misclassification Analysis of Model Responses

While it is important to critique NLP models, we must also challenge the automatic assumption that these models are always inaccurate and consider the relative performance. We initiate our fine-grained analysis beyond accuracy by analyzing which labels were commonly mislabeled for both sentiment and emotion classification.

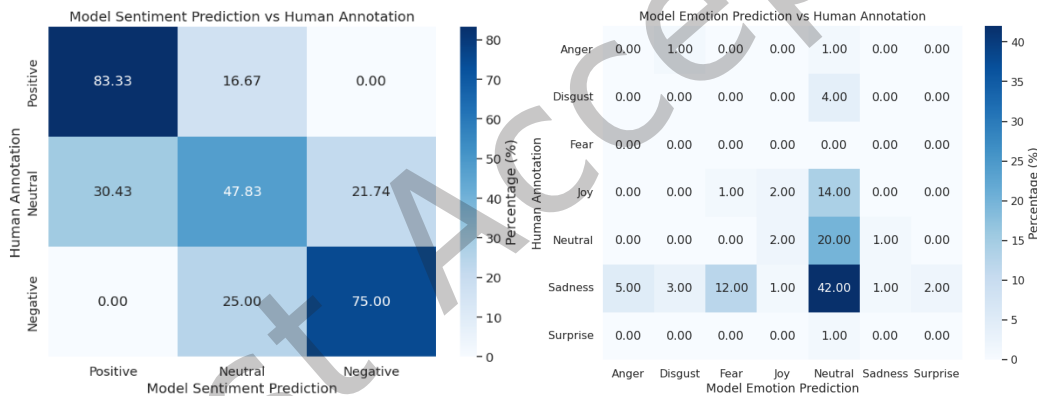


Figure 9: Confusion Matrix of Model Predicted vs. Human Annotated (a) Sentiments and (b) Emotions

According to Figure 9(a) we observe that model predictions for the positive and negative sentiment labels aligns well with human annotation (positive – 83.33%, negative 75%), while annotators considered many neutral class predictions as either positive or negative. This signals that the model failed to pick the nuances of the sentiment in the responses it identified as neutral. In Figure 9(b) we compare the emotion categories for model responses against human annotation and find that the most common case of mislabeling was considering sadness as neutral (42%) and joy as neutral (14%), showing that the model failed to understand the deeper emotional context. Both models mainly show mistakes in identifying neutral categories. A key reason behind this discrepancy lies in the differences in annotation targets and model training datasets. Sentiment classification is inherently coarser, distilled into three broad categories—positive, negative, and neutral—allowing for more generalizable predictions across datasets. In contrast, emotion detection aims to assign specific psychological states (e.g., fear, joy, sadness), which are not only more contextually nuanced but also deeply embedded in cultural, social, and situational cues that general-purpose models often fail to capture. On top of that even people often disagree in assigning emotional labels [133]. We expand our analysis with an audit of the pretraining data in the next section.

5.4.3 Pretraining Data Audit of Models

Upon a review of the pretraining and finetuning data sources used for both models we found that while the sentiment model was trained and fine-tuned on only Twitter-based conversations [82, 83], mirroring the informal and expressive tone of our participant quotes, the emotion model was trained on combination datasets like Meld (collection of dialogues from a popular TV show called Friends) [86], and GoEmotions (Reddit dataset, western, young male centric demography) [87] along with Tweets from Twitter [139, 140] and emotion survey data curated by psychologists [141]. The mismatch of data for emotions may account for the performance issue, both Meld [86] and GoEmotions [87] contain brief, decontextualized, or media-centric samples considering western culture more, so these datasets may not reflect the complex narrative structure and emotional layering of participant letters written in response to gendered personas. Emotion model may have predicted "neutral" where humans detect "sadness" because it struggled with contextualization, especially in emotionally ambiguous or indirectly expressed content. Furthermore, gendered language use and local expressions of vulnerability (e.g., expressions of collective struggle or institutional barriers) may not have been captured properly.

5.4.4 Reflecting on Power in Data, Tools, and Knowledge

Inspired by bias-aware "Situated Data, Situated Systems" frameworks for examining power relations in NLP [134], we conclude with reflections on four key questions that surface the power dynamics embedded in our methodology:

(1) Who or what is included in the research?

Our study centers first-year women computer science students from four Bangladeshi universities who wrote personal letters to two personas: M3, a dismissive male figure, and Oporajita, a supportive female presence. We also include sentiment and emotion analysis models commonly used in English-language NLP research.

(2) Who or what is excluded?

The models are trained on English-only, Western-centric corpora and emotional taxonomies (e.g., Ekman's six emotions), which may not reflect South Asian linguistic and cultural contexts. [135, 136, 137] They also fail to capture the nuance of letters, affective storytelling, or gendered political expression.

(3) How is knowledge defined?

We treat the letters as situated expressions of emotion and critique—not abstract data. Studies show that annotator demographics significantly shape sentiment labels, affecting model performance by over 4.5%. This highlights that what models "understand" as sentiment reflects whose emotions are centered in training data. Our findings expose how NLP tools often misalign with the lived realities of our participants, making model validity not just a technical issue, but a matter of epistemic justice [138].

(4) Who has agency and who is empowered?

Participants exercise partial agency within structural constraints; researchers hold more power in interpreting, modeling, and framing. While we initially used NLP tools as visual aids, our error analysis revealed systemic mismatches that called for deeper critique.

Prior work highlights the importance of accounting for annotator demographics when building and evaluating datasets [138]. The emotion model we used reported 66% accuracy on its original test set, but this dropped to 21% when applied to our dataset. This stark decline reflects more than a performance issue—it reveals a demographic and cultural mismatch between the model's original annotators and our context. Our Bangladeshi annotators bring situated knowledge shaped by lived experiences that Western annotators likely do not share. If the same letters had been labeled by a different demographic, the emotional readings—and thus the model's "failures"—may have looked very different. As feminist HCI scholars, we argue that what counts as a failure is

not objective; it is shaped by who is looking, and what assumptions they carry. Tools do not just fail—they fail someone, and how that failure is named matters. We advocate for *context-aware dataset construction* and *evaluation* along with developing *culturally sensitive models* that centers participant realities and community validity— when situating tools in a particular culture such as ours. Even tools like NLP models must be co-constructed with the communities we aim to serve.

6 PARTICIPANT DESIGNED SOLUTIONS AND RECOMMENDATIONS

In this section we outline the recommendations and solutions designed by the participants in the co-design workshops we held. The participants designed a variety of solutions for finding support in their learning environment, family and society and transportation sector.

6.1 Support in the Learning Environment

Gender sensitive and aware teachers, staff and facilities along with opportunities to collaborate in groups for further enhancements were discussed through letters written during fictional inquiry sessions and solution design sessions by the participants.

6.1.1 Gender Sensitive and Aware Teacher and Staff.

The necessity and the impact of gender sensitive teachers and staff was an area that clearly came out from the series of workshops. This topic was not explicitly shared in other studies focusing on women in computing in this region [3-7].

Most participants across all the regions strongly referred to the importance and impact of having gender sensitive teachers and staff. The topic was intensely discussed in the Rajshahi and Sylhet region. The solution ideas preceded the points raised through the fictional inquiry sessions and follow-up discussions. In the letters written to Oporajita, participants shared faculty members' negative attitudes towards female students. The negative mindset came in two forms, one referring to the attitude that women lack technical ability, and the other one was the lack of necessity for technical knowledge as women would not be entering the technical job force. One participant strongly shared her views:

“It would be wrong to say that the gender stereotype in this discipline still lies in my university, yet it hasn't completely gone away. I still have teachers in my classroom who underestimate my ability in lab work. And, on most days I am tired of being labeled as a *female* engineer, not just an Engineer.”- P38, Chattogram.

In the design workshops, the solution approaches came where gender sensitivity of faculty and staff members reappeared through design suggestions. It came in various formats referring to short-term or long-term exposure of the teachers and staff who have a strong impact on their students. A participant P20 suggested “*gender sensitivity test prior to a faculty hiring process*” referring to the tertiary level while another participant P15 suggested “*counseling on gender sensitivity at the primary school level*”. Another participant boldly shared her desired design process as follows:

“Counseling sessions and mental fitness testing for teachers regarding their mentality problem. They should treat students equally regardless of gender. [...] Be flexible to students so that they can communicate without hesitation.” - P25, Sylhet.

The gender sensitive role requires an active engagement in the entire learning process as participants shared. Our participants elaborated on ongoing challenges in congested lab spaces or when few resources are shared among groups of students during hands-on learning activities. This was explained by a participant during the design phase:

“In our 1st/2nd semester, we completed some labs based on electrical experiments. We faced unequal opportunities for advancements or leadership roles during those labs. Most of the time, all male

students successfully did their experiments with the help of the teacher. But we, females, failed to do our lab as we are a minority group. During those experiments, all male students made a crowd and completed their experiments as a team. But we are a minority team, failed to enter that crowd, failed to hear the teacher lecture and so on.”-P43, Chattogram.

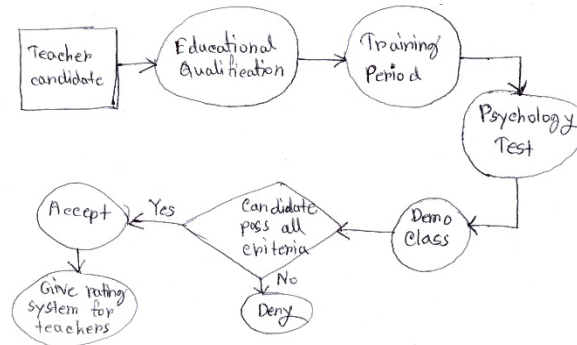


Figure 10. A flow chart showing the proposed teacher hiring mechanism

The design workshop had several lists and written formats along with one elaboration of a teacher training process. In the illustration, the teacher training process initiates at an early level - the participant mentioned initiating training from primary education level as presented in Figure 10. The design illustration refers to a psychology test about a teacher’s mental preparedness to teach gender equality followed by a demonstration class to check it - a topic that was seriously discussed by many participants.

However, the disengagement of male teachers and staff is sometimes intentional, justified by religious regulations as the study showed. One participant P15 shared it through her letter in Oporajita as: “Some teachers behave weirdly with girls. Some religious teachers don’t want to communicate with girls in the class. This is true that for religious purposes, it’s fine. But we went there to study. If teachers don’t want to communicate with us, it will be tough for us to study there.” The religious value-based separation also affects the group learning where male peers are not open to engage with their female classmates. A participant P47 shared: “Due to religious attitudes or social norms, many boys do not want to participate properly in class activities with classmates of the opposite sex. In this case, various group activities or discussions are hindered, and girls lag behind.” The isolation for strict religious practices was a concern for the learners also; participant P42 shared: “Limited representation restricts my visibility, voice, and overall influence in these fields.”

The participants directly associated the mindset of teaching and staff members towards the growth and dreams of female students referring to the very low number of women in tech-industry. The number of references to gender sensitive teaching staff in the problem identification phases was over twenty-four references by seventeen individual participants as well as in the design phase by thirteen individual participants across various regions which shows the importance of this problem.

6.1.2 Facilitation through Wider Exposure and Better Infrastructure.

The increased exposure and learning facility in technology was widely discussed by participants from all the regions. The topic of better learning facilitation came from two perspectives - prioritizing and facilitating Information and Communication Technology (ICT) education throughout the country and focusing on specific infrastructural issues women face. The need for inclusion of ICT at a wider level with increased rigor was discussed equally through letters to M3 and Oporajita along with the design sessions. In the second topic discussion, there were considerations about female learners that were relevant in residential universities outside the capital which came through the design session.

ICT is given a higher priority at country level where it is included as a subject at high school level in Bangladesh [8]. Study participants have also emphasized the quality of education and rigor of ICT learning from younger age as a participant P44 mentions: “Tech competition in higher school level or basic programming contest at higher school level as ICT subject is introduced at school level.” However, the students experience ICT knowledge gap for not having enough qualified teachers in this field as P32 mentions: “To ensure proper teaching

of ICT, teachers have to be students of IT background.” The broader discussion remained about having wider exposure of ICT in major curriculum such as the National University Curriculum to ensure access of ICT to the root level students according to the participant. If more people graduate from ICT, it will open up opportunities for a wider range of students along with better job creation in this sector, and subsequently creating more qualified teachers in this sector.

There was discussion on addressing current problems in the learning environments that are relevant for female residential students. The public universities in our studies generally are located outside of Dhaka in remote regions and host female students in residential halls. These female halls are often located in secluded regions to increase their security. Female students must return to their residential halls by sunset by institutional policies which restrict access to laboratories. A participant P30 shared: *“Restrictions to access the lab after a certain time led to losing interest.”*

Beyond the policy level restrictions, the students in the residential halls also stated the need for better internet connectivity. The current challenges in ICT learning were discussed seriously by several engaged participants. The remote locations of female halls to ensure safety restrict access to the availability of mobile internet data from those hostels. P32 shared: *“Sometimes the mobile data doesn’t work properly here.”* It also has related disengagement as P14 shared: *“There is no proper network in my hall. In fact, due to this issue, I stopped many online trainings”* which may propagate through their further journeys.

In the design session, P25 stated the importance of *“Good internet connection in female halls.”* and participant P14 shared the impact of having limited internet access, *“In my hall, there are utmost 2000 female students. But we have just one internet room. Many of us don’t even know how to operate computers. It’s because they don’t have the opportunity to learn.”* A clear design solution came from another participant:

“Make a lab for IT students or non-IT students inside the hall or hostel where they can help us in our lab. Internet speed in the lab is an important issue. In Sylhet, because of the rain the internet is bad. If we have fiber internet it will be very helpful for us.” - P43, Sylhet.

6.1.3 Facilitation through Mentoring and Role Model.

Many participants emphasized the importance of collaboration through networking and mentorship in our design workshops and suggested technological solutions for community engagement and addressing underrepresentation of women in every sector from educational institutions to workplace. A number of participants (sixteen) referred to collaborative frameworks as a solution approach through the design session. The participants clearly referred to underrepresentation of women in their letters to M3. They shared concrete ideas on networking where P43 shared, *“Female Teammates Can be a Savior”*. P45 and P2 asked for *“support network”* and *“student forum”* respectively. As a design solution, P48 and P19 referred to the idea of *“website for networking”*. It was illustrated with more details:

“An online platform to collaborate between girls in the same department as the student ratio is low. There can be a website where students can ask questions and female mentors will reply.” - P32, Sylhet.

Moreover, mentoring support from professionals, role models, peers were shared as a design solution by as many as twelve participants. A participant P29 asked for *female mentorship to ease up entrance into the IT industry*, P38 asked for mentorship from *successful ones* while others referred to mentorship in general. Another participant addresses the requirements clearly as: *“I think for solving these issues, we need some mentors. They can conduct sessions in our university. and give us proper directions. Also, any event should be published properly so that students can get to know about the event and can participate.”* - P14, Rajshahi. The participants shared information gaps among female students about ongoing opportunities for the lack of a larger network among women compared to their male peers.

It shows that the tertiary level female students are aware of the possible solutions around the academic challenges through their direct or perceived experiences on existing challenges for women in computing.

6.2 Support from Family and Society

6.2.1 Parental Preparedness for Gender Sensitive Upbringing.

The gender sensitive parenting as a key solution came out among many during the design session while the challenges it created came out through the letters towards Oporajita and M3. The discussion on parents was not discussed in previous studies exploring the challenges in the context of Bangladesh [3-7], showing how the fictional inquiry allowed participants to talk about challenges that are coming from intimate family members. We present the challenges followed by the solution approaches participants shared in detail.

The study showed how lack of gender sensitivity limits opportunities for tertiary level female students. The limitations vary from parental influence in making decisions about subject choices (e.g., engineering or not, computer science or not etc.); career choices as female professionals or parental decision on deciding the time of marriage. A participant shared frankly about subject choices in her letter to M3: *“Some families insist their female children not to study Engineering. They always say that Engineering is the field for male. Girls will face limitations in the job field if they choose engineering as a career. Field jobs for women will be full of problems and insecurities. As a result, females become frustrated and do not want to study engineering.”* Apart from subject choices, a number of participants referred to parents pushing their daughters towards getting married before graduation. For example, a participant shared how many of her female classmates are not able to continue their studies as follows:

“As a 9th semester student, I saw a lot of my female friends getting married, they can’t even complete their studies. It’s not fair, right? In this society girls don’t get enough opportunities in terms of jobs. I hope every family tries to treat their girls as humans not as a burden to get off. I’m doing my best to get my equal share of responsibilities in society.” - P12, Rajshahi.

As solution approaches to address parental preparedness, participants shared various innovative ways during the design workshop. It had a range of solutions starting from counseling and psychological tests through applications to support parents to provide adequate support for their daughters. Some of the participants focused on the importance of the topic to be addressed during the pre-parental or early parenting process. For example, participant P14 shared about a possible technology solution through: *“an application where parents can find out how to spend time with kids, and the importance of teaching equality to their kids.”* Another wished for couples to prepare before their parenthood as follows:

“Couple who wants to be parents: Before being parents a couple should learn how they can be ideal parents and how they should raise their children. They should learn: - anger management, how to control temper, and how to keep patience.” - P13, Rajshahi.

Some participants focused on solution approaches for current parents. A participant P20 shared: *“We can make a counseling group for parents which can motivate the parents about equal importance of both male and female students.”* While some participants exhibited a positive attitude towards suggesting ideas, there were participants who accepted the current situation. For example, P27 shared, *“Unfortunately, changing this societal perspective is a gradual process that requires time and effort. So, I’ve come to accept that changing their mentality is beyond my control. Instead, I must focus on coping with it”.*

Figure 11 (a) and (b) shows design visualization developed by participants during the design workshop where one refers to parental preparedness while the other talks about the points to be maintained in the perfect world. The discussion about the perfect world involved the educational sector, awareness and implementation of gender-neutral policies in various places (e.g., workplaces).

It showed the importance of parental support from the viewpoint of the participants from an early age towards reaching one’s career.

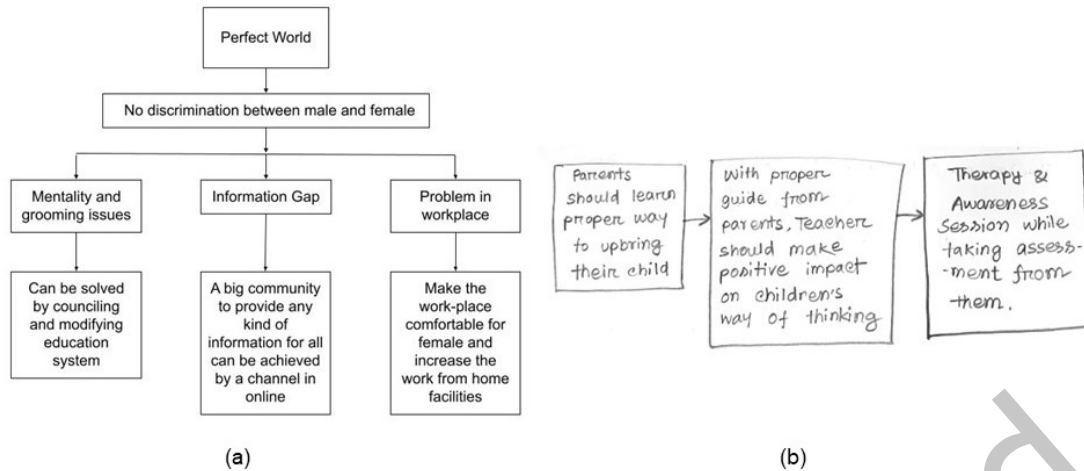


Figure 11. (a) Elements required for a perfect world (b) Solution to ensure parental awareness

6.2.2 Facilities and Awareness on IT and IT Job Sector.

The role of a broader view from societal support requiring increased levels of awareness to assist women in computing was discussed by the participants where it came out during the design sessions.

A strong discussion came about societal bias and ways to address them to enhance societal acceptance and support towards women in computing. A participant P15 shared about existing negative social bias towards working women in general in her letter to M3 as: *"Even there are some people who believe that women who work outside home are the culprits and damaging the society. People think that women working outside makes their home messy, makes society decreasing the status."* There is a regional existing influence on societal preferences of women's jobs. As a participant P24 shares: *"Media highlighted female's role as doctor and teachers which should be changed."* A participant shared about her own journey during the design session about addressing societal biases:

"Speaking for myself, one of the challenges I encountered was getting into an Engineering University. My father wanted me to become a doctor because, according to our society, it was considered more suitable for girls. However, I wanted to explore myself as an engineering student. I had to endure the disapproval of many relatives, and I was afraid of potential embarrassment if I didn't get accepted. But with sheer determination and a stroke of fate, I made it here, and I'm happy with how things worked out." - P44, Chattogram.

While the majority of the participants are unmarried and currently students, based on their perceptions about the workplace of Bangladesh's IT industry, they have asked for enhanced facilities like practices in gender-sensitive technical companies abroad. There were discussions on paternity leave along with existing maternity leaves by participants. For example, P31 asked for *"family-friendly policies such as maternity and paternity leave and flexible working hours that can enable women to balance their personal and professional lives effectively"*. A participant P42 asked for *"Newborn baby caring services, babysitter services to manage household"* and P44 asked support system in more detail as follows:

"Increase online facilities among working mothers. Enforce corporate houses to facilitate working mother as they have children at home. The government should fund IT Companies in remote villages to make the job field more accessible to women. Advertisements in tv and social media for awareness." - P44, Chattogram.

In contrast, there was a strong thread of discussion on creating positive societal views toward women in computing with the help of social media, print media, movies, cartoons, and publicity. The participants shared current trends in motivating societal changes as P20 shared in her design session involving: *"Online influencer*

for career motivation.” Participant P42 shared in her design session: “Build awareness by advertisement, different types of online TV shows to show how women can be successful in tech.” and P38 shared: “people need to publish positive achievements of women in technology in news”.

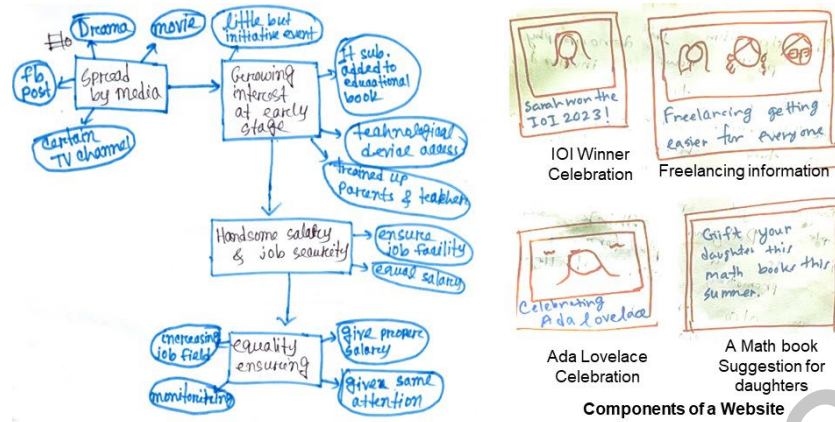


Figure 12. (a) A comprehensive solution to support women (b) Awareness generation highlighting women.

The participants shared their views on ways to increase awareness, visibility, and eventually support systems for women through their illustrations as presented in Figure 12. Figure 12-(a) shares an entire ecosystem to support women while Figure 12-(b) shares how women’s achievements should be shared in various media outlets. Moreover, there has been discussion on increasing societal awareness that creates a positive image on young minds in a similar way the cartoon character Meena has made a positive impression on girls’ education during the last decade in Bangladesh [2]. A participant suggested Meena should now have her computer as shared in Figure 13-(a) while it used to be Meena going to school previously as can be seen in Figure 13-(b).

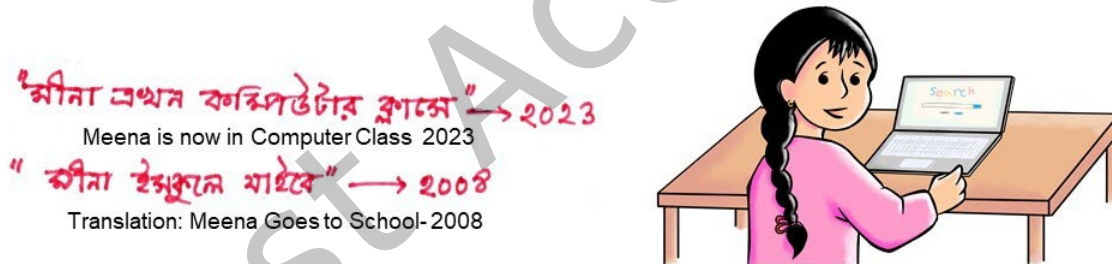


Figure 13. (a) Comment on Meena going to computer class (b) Meena using computer in school.

The design session participants shared time and effort to think about ways to improve the visibility and acceptability of women in computing showing its importance and necessity to them.

6.3 Support towards Safe Transportation Facilities

Gender-sensitive and aware teachers, staff, and facilities along with opportunities to collaborate in groups for further enhancements were discussed through letters written during fictional inquiry sessions and solution design sessions by the participants.

The barriers related to safety were discussed through the letters. A participant P5 shared the challenges of safety in her letter to M3 and how it impacts mobility and parental concerns for their children: “In our country, there are some safety issues for women. I think this is the main barrier for the women and that’s why families don’t want their children to go outside to work.” The safety concerns are closely coupled with the lack of safe transportation facilities since there have been ongoing incidents of abuse and harassment in various public transportation where women have been the major victims [47, 48]. Unsafe transportation as a challenge for

women and relevant solutions both came from the design workshop explorations. The challenges were discussed during the design solution to express the barrier it creates for participants. A participant shared:

“I have participated in many inter-university hackathons and programming competitions. Each time, I faced the problems of the night journey as a female since in daytime we have classes, exams, etc. Especially when I need to go by night buses in Dhaka there is a big security issue if I went alone. I think these security issues also discourage many female students from participating in inter-university competitions.” - P33, Sylhet.

The transportation related solutions varied widely and provided innovation through technologies where some used existing ideas of having a dedicated emergency response team, website/app for security along with innovative ideas with emerging concepts. A participant P45 clarified the requirements as: *“Dedicated transit for women to safely get to work”*; P31 asked for: *“Available seats for females in any transport. Available transport in rural areas. Female Uber or Pathao (ride sharing) driver.”* and P15 mentioned: *“Application based bus service should be implemented. Nearby empty Seat tracking by application service.”* while P33 asks for an AI-based solution: *“Registered transport should have CC-TV cameras integrated with AI technology and GPS. AI will detect any occurrence and send a notification to the security department (Police, Rab, or dedicated call center).”* Some participants had a vivid picture of a solution approach:

“Increase surveillance in public transport. Track the transport as the transport will be registered in the government application. The application will be featured with transport location. CC camera, emergency button, and location tracker in the public buses. After pressing the emergency button, it sends the signal to the central through the tracker and sends the notification to the nearest station. - P27, Sylhet.

Moreover, participants discussed in great detail about possible technology solution approaches where safety can be ensured through seeking help from nearby Police stations, designing innovative transportation solutions and ensuring a safe environment as presented in Figures 14. The solution on Figure 14-(a) shares an integrated system that allows one to file a complaint where the system is connected to law-enforcement systems, portraying Governmental rules and regulations. Similarly, Figure 14-(b) refers to a system that is described in writing where one possible victim can get series of support depending on the location, time and intensity of the problem one faces.

The expectations as a citizen to live with liberty of being safe and respected is shared in Figure 13. Figure 15-(a) shows a girl walking at night alone without facing any safety problems, Figure 15-(b) shows a girl walking with a friend at night and Figure 15-(c) shows a group of girls enjoying a concert without fear of harassment.

The participant’s engagement in the discussion about the desire to have a safe environment shows the lack of safety as a major ongoing problem for women in this region.

7 DISCUSSION

This work engaged tertiary-level computer science students as the future representatives of the women in the computing field through Fictional Inquiry (FI) which has been rigorously illustrated with Natural language processing (NLP) methods and relevant visualizations followed by the co-design workshops. The goals of these methods were to bring out their challenges and solution approaches respectively.

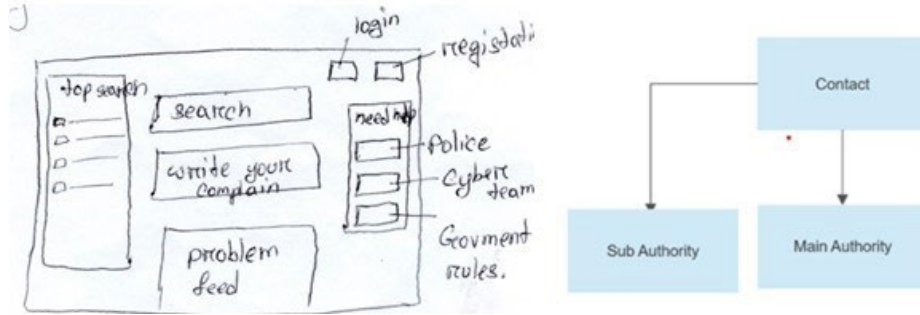


Figure 14. (a) A Website that provides safety (b) A safety application design involving various authorities.

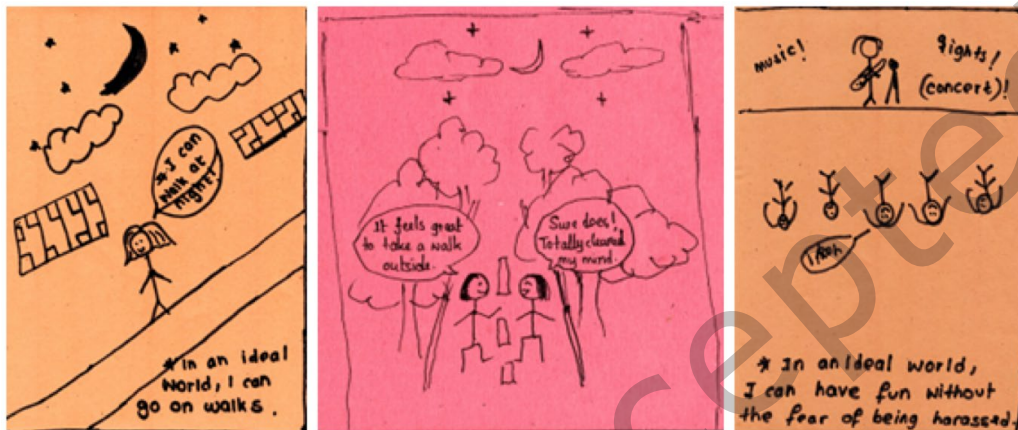


Figure 15. An ideal world where women walk safely at night (a) A women walking alone at night (b) Two women walking and chatting at night (c) Women going to a public gathering such as a concert without being harassed

7.1 Critical Reflection on the Study

Navigating Power Dynamics during Workshops: Following the work of Harrington et al. [132], in this section we reflect on the methodology and participant engagement of our study. The diverse composition of our research team, encompassing various gender, religious, and geographical backgrounds, has been instrumental in fostering a holistic approach to understanding the experiences of women in computer science education in Bangladesh. Given our shared yet varied perspectives, we recognized the need for a methodological design that minimized power dynamics and encouraged honest reflections. By incorporating indirect inquiry through fictional personas, we facilitated an environment where participants felt comfortable articulating their experiences and concerns without fear of judgment or repercussions. Our familiarity with local socio-cultural dynamics allowed us to identify subtle cues and implicit concerns in participant narratives, adding depth to the insights gleaned from their letters. Being from the same environment allowed us to create a supportive and empathetic research setting, which facilitated the participants' openness in expressing their challenges and aspirations within Bangladesh's unique context. By using Fictional Inquiry (FI) and indirect engagement with personas, we minimized direct power imbalances, allowing the participants to envision solutions free from potential social constraints.

Empowerment Through Voice and Imagination: The question arises whether the usage of open design opportunity provides a *blue-sky solution-centric* concept, which refers to 'thinking not grounded or in touch with the way things are' [124]. A solution design or thinking of the world one wishes to operate in, may appear as a heavy one where one may not have much power to make such positive changes, and this thinking can be disempowering from one perspective. Considering technology as a solution has also been criticized as being a euphemism [125], where in this work, the thinking takes place around some solution design. However, we have considered the open page provided in FI as a way to share one's voice. The participants engagement in this research reflected their positive engagement in the exploration process. The participants referred to the *flaws*

in parenting methods and teaching biases in Computer Science and Engineering classes, which were never reported in prior studies considering women in computing in Bangladesh [3, 4, 5, 6, 7]. A particular source of motivation to allow an open thinking space came from the long experience of the first author's work on addressing the problem of sexual harassment, where women are the major victims [63, 127, 128]. The social problem of sexual harassment has remained, and in many contexts has worsened, spreading from physical space to cyberspace, where it spreads rapidly. A solution design approach to support women to avoid or handle a possible attack may seem useless apparently from outside since the problem remained. However, when the work was paused, and the researchers were not sure about the possible impact of this design, the participants approached the researchers and shared how the discussion gave them strength, hope, and validity [127, 129]. The researchers engaged in the design process shared how being part of this project made them feel that they were doing something important. The problem remained, so did the solution design and the discussions around the design process worked as the activism against a heinous social problem [127, 129]. The social and structural problems coming from lack of safety, security, and infrastructural support require large-scale investment and prioritization where policymakers and authorities' involvement is needed, a place where there is still a gender imbalance [122]. However, the current limitation should not be a barrier to dreaming of a better environment where the thought process is liberating, which is far from the dreams of many others from past generations or who are in different situations. There are women in computing from Bangladesh who are able to enjoy a safe environment pursuing higher education in foreign regions, as a recent study has shown [123], a dream that has probably stemmed from seeking change. As researchers, we want to keep the conversations going, even if it feels hard to achieve, since this discussion is better than silence, which allows problems to persist without being challenged. Thus, the design process came from opening up a conversation, thinking about a design solution as an enabler to break the silence that usually remains when it comes to women's problems.

Contextualizing the WiC Participants' Engagement: While prior work [117] found tensions around the activity structure of participatory design workshops with marginalized communities and participants indicated that “*brainstorming ‘out-of-the-box’ solutions*” were not moving towards actual solutions that can be implemented in near future, our participants did not show any similar behavior. In contrary, the participants were engaged with the process and many of them wanted to reach their vision of the world for at some point in various ways suggested for personal, family level, social or policy level improvements. There could be two possible reasons for this variation. **First**, women in computing (WiC) in our study experienced marginalization primarily based on gender, which differs from other marginalized communities that often face a combination of financial, educational, and gender-based barriers [14, 15]. Many communities struggling with technology access encounter intersecting challenges, including economic constraints and limited educational opportunities. In contrast, the women in computing who participated in this study largely did not report financial or educational marginalization. This may be partly because they have already demonstrated competitive academic success to gain admission to educational programs, with family support that some other women in similar age groups might lack, such as those who leave school early due to social pressures like early marriage [18]. All the public and private universities where we conducted our studies are highly competitive and prestigious. These women who participated in this study are in a trajectory to enter the professional field and may contribute to designing changes for future cohorts. Their existing familiarity and access to technology could explain their interest in envisioning technology enabled solutions. **Second**, prior research on Bangladeshi women showed that having a voice to express what one wants and the ability or freedom to dream of a better world can be considered as an opportunity in our context. Since many participants were from computer science background, many of them preferred to ideate technical solutions such as suggesting apps for parents and teachers. Prior studies engaging the guardians accompanying women in computing to attend national level programming contests and international robotics competitions show that many parents, especially mothers, were not allowed to dream and thus, they wished to make the paths easier for their daughters [3, 4, 5, 6, 7, 130]. Many of these parents provided both emotional and financial support. Reflecting on their personal difficulties was a motivator to work towards thinking of ways that can be useful for their daughters. In Afghanistan, a similar phenomenon was observed among women in computing who were refugees and returnees themselves and worked to support their junior ones to have easier access to resources prior to the Taliban Government's takeover [40]. Naila Kabeer's work showed how a group of women have invested in their daughters to ensure they are well educated

and able to earn, ensuring an enabling environment for their daughters, one that was not present for themselves [6].

Researcher Positionality and Shared Context as Interpretive Tool: Our positionality as researchers who have navigated similar educational structures equipped us with a unique lens for interpreting the data. Our reflective approach in this study also acknowledges the dual role of our identities as both insiders and facilitators. Navigating this space required us to consider how our shared experiences and social conditioning within the same educational and cultural framework might subtly shape our interpretations. The balanced dynamics within the research team helped us remain reflexive throughout the process, allowing us to acknowledge and address any potential biases that might arise from our shared backgrounds. By actively reflecting on our positionality, we aimed to strike a balance between empathy and objectivity, ensuring that the participants' voices were central to our analysis and that our interpretations were grounded in their lived experiences rather than our own assumptions. Our choice of method reflects a commitment to respecting the participants' agency, especially in a setting where critique of authoritative figures is uncommon. We also recognize the empowering potential of our research activities themselves, which may have offered participants a rare opportunity to voice and validate their experiences and solutions. Additionally, as researchers familiar with the lived realities of Bangladeshi women in computing, we were uniquely positioned to interpret and contextualize their design solutions as deeply rooted responses to structural and cultural challenges they face. This positionality not only enriched our analysis but also underscored the importance of reflexivity in fostering authentic engagement and advocating for meaningful technological futures for women in Bangladesh. For example, the ideal world where a girl can freely walk at night, may initially show a complete detachment from the world of computing, but the researchers could easily vision themselves walking from a late-night programming class without the worry of getting harassed while riding a public transportation in Bangladesh. Our shared background allowed us to recognize specific cultural and infrastructural challenges that might otherwise go unnoticed by external researchers, such as the intricate dynamics of gender-sensitive teaching and societal expectations surrounding security and mobility for women. Suggesting a future Bangladesh where teachers and parents are going through psychological tests may seem initially far-fetched as an idea, but our participants are coming from a generation of social media usage where continuous activism and community engagement do result in real, concrete changes. Acid throwing, a violent crime where women's faces were often disfigured via throwing acid by perpetrators and 80% of the victims were women, came to a rapid decrease in the past years in Bangladesh (a drop from 300-400 attacks per year to around 30 crimes per year for the last six years), as a direct result of legal changes and awareness generation campaigns via non-profit and government campaigns [118]. Since the Bangladeshi participants have seen development from their early years where large-scale social changes are possible when everyone in society works together, they attempted to use our research study as a platform for advocating their voices.

Reflexivity in NLP Analysis: Our use of NLP techniques to analyze responses added a structured, data-driven layer to the qualitative insights, enabling us to capture sentiment variations and compare thematic patterns across different sites. We also approached this analysis with sensitivity to the complexities of lived experiences, acknowledging the limitations inherent in sentiment analysis and the need for nuanced interpretation in such culturally rich and emotionally complex topics. The data visualizations show us snapshots of participants feelings where they were comfortable sharing their innermost feelings with a fictional character Oporajita, suggesting perhaps the participants want a real life Oporajita in this generation, just as her fictional ancestor Begum Rokeya proudly carved the path of feminism in Bangladesh.

Contextual Design for Student Agency : In prior work [121], it has been suggested that “**‘poverty of the mind’ prevents people from embarking on constructive activities**” and marginalized communities can choose to “**accept and confront their situation and control their own lives by doing something constructive with whatever few assets they had**”. Drawing parallel to this, when we, a group of students and researchers from South Asia, decide to ask our own junior peers about the challenges they face through FI and utilize their own sentiments as visualizations to draw the actual feelings of fear or disgust, however taboo they may be, against their and our own authority figures it shows a form of agency to utilize all current tools available in the scientific communities disposal to create policy and technological solutions against the problem of underrepresentation in computing. It also opens the path to collect feedback from ICTD and HCI community

to iterate and improve upon the bare-bone solutions students provided. Although designing for ideal world for women in computing in our co-design workshops might lead to the perception that the students are asking for a world free from all biases or social prejudices, in reality our students asked for very simple requirements such as the safety to walk at night on their own and safe transportation facilities to go to schools. These are their deepest and innermost feelings about the current circumstances in Bangladesh, which the policymakers have not yet prioritized. However, we as researchers present their vision as it is in our findings with our ethical obligations. In a future work, utilizing ‘think-aloud’ strategy [119] where the students would talk about their solution steps aloud may help to understand their thought process with refined granularity. Also, the prior work showed that in collectivist societies such as Bangladesh, group studies help to talk about taboo topics [120], thus we would probably utilize a group setting such as the current workshops.

7.2 Practical Relevance of the Findings

The in-depth analysis of FI from qualitative methods, illustrated by Natural language processing (NLP) shows the myriads of emotions and sentiments participants shared in their letters. It showed that the sentiments conveyed in the letters were diverse and emotions such as sadness and anger were present. During FI participants wrote letters to fictional characters without fear which helped them to express their concerns in a safe space. Notably, FI has never been used in women in computing context. While prior work showed the importance of supportive family and teachers [5-7,18], in our work we found stronger emotional responses related to the lack of support from intimate family members and teachers and advocacy for novel solutions such as counseling and psychological tests for both parents and teachers. It clearly shows the lack of safe space for women to share the discriminatory behaviors they have faced in their homes as well as in educational institutes. It requires a broader level of awareness at family-level, institution-level and policy level to ensure gender-sensitive learning environment. The religious justification of many gendered behaviors has long term impact on the learners, and it must be addressed at higher levels. The strong feelings coming from letters of the young WiC who will enter the job sector soon illustrated by our NLP analysis of sentiments and emotions show the need to address them urgently. It works as a motivation towards individual, institutional and policy level interventions to address existing challenges.

The co-design method allowed the participants to engage actively to address the problems seeking their desired changes. The co-design solutions participants asked for reflected a lot about the current situation and status of women in computing in Bangladesh. The set of solutions seeking gender sensitivity among parents and teachers shows the bruises participants have endured from their previous experiences. This discussion required courage and shield provided by the FI method considering the close-knit collectivistic society along with its patriarchal practices [50, 69]; as speaking up against parents and teachers can be considered taboo in the context of Bangladesh. The designs participants provided may appear to be simple and direct, but the students honestly raised voices on what they truly want without any pretense. We acknowledge that the long running societal problems related to WiC are not easy to address and possibly not through a half-day workshop. However, the workshops are a starting point and the design contributions and optimism of the young WiC shows how eager they are to have many of these problems addressed. It also varies from the solutions provided in existing contextual literature [3, 5, 7, 10, 18, 110]. Existing literature focused mainly on gender sensitive policies and awareness generation. While the students in this study also had similar suggestions they also advocated for major cultural and mindset shifts. They asked for gender neutral parenting in a cultural context where parents have strong authority over the lives of the children and often make decisions on behalf of them [113]. A psychological test for parents may appear trivial from outside but it is coming from the hope to ensure that the parents are ready to raise their daughters with strong values. Moreover, they asked for gender sensitive teaching and continuous evaluation of the teacher’s values that would not affect their students. Such suggestions coming from participant’s lived experiences cannot be ignored.

The strongest solutions came from the participants regarding technology to support safe transportation - an ongoing problem that appears to be extremely important to the participants, showing the continuing insecurities women face in Bangladesh [47, 48]. Transportation and security problems have been referred to considering attendance of computing related events and workshops in different literature in prior work [5, 47, 48]. As our participants are eager to make progress, it is frustrating to see challenges such as lack of safe

transportation remaining as a problem for decades without any major action taken by the Governing authorities. Moreover, other infrastructural problems such as lack of access to labs, difficulty to avail internet etc. have been mentioned before as a barrier and have not been addressed till now [5, 7, 18]. These problems are comparatively straightforward to consider if adequate priority and attention are given to them. Thus, there is a requirement for a gender sensitive support system for women, enforced at the highest level. Overall, the solutions, through existing barriers, reflect the current standing of these women in computing. The question remains why these problems are so long standing and what can be done about them.

Currently, the gender sensitive education policy at the primary and secondary level has shown great success and progress among female students [68]. It needs to be propagated further with the help of supportive and required stakeholders. In our study, we have observed that the tertiary level students in computing are eager to propose solutions to existing problems. The solutions require multifaceted approaches from parents, teachers, society members, and adequate authorities. Novel solutions explored by co-design workshops such as the visual representation of Meena working on her computer can spark the dream of a future woman in computing. As a solution, gender-sensitive discussions should continue to ensure the simplest dream of “walking safely back home from work (read IT Office) on a moonlit night” to come true for any little girl growing up in Bangladesh.

7.3 Relevance of the Research in Current Time

The exploration around the relevance of this work will allow us to understand how this particular exploration enables the women in computing community, stakeholders, policy makers, researchers which in turn contributes to the society.

The barriers women in computing face have been present from the very early years of computing [111]. Many of the structural barriers have been addressed while the social barriers have remained [64]. The number of women in computing remained low even in gender-neutral countries like Norway [64, 107]. It shows how the barriers are deeply rooted and high level gender-neutrality index is not able to address it. On the other hand, the number of women were higher in gender restrictive countries like Saudi Arabia considering it as a job where women can contribute from home [64]. Here the higher number of women in the field contributes to limited mobility aligned by societal barriers. Thus, the gender neutrality index or the number of women in computing does not tell us the entire story.

The societal problems are difficult to address as they are context sensitive in nature. It varies in regions and evolves over time. In a South Asian country, a woman in computing faces social barriers that are localized and often is not noticeable by the victims for its long-term existence. A tertiary level student in our study, a woman in computing, shared after a discussion with others that she did not realize that many of the existing practices were barriers. Along with its invisibility, their barriers evolve over time. In 2024, women are not facing barriers to acquire basic education, like the battle Begom Rokeya had to fight [77]. The barriers women face today are more subtle. They are facing subtle disrespect from teachers and peers or having narrower subject choices guided by parents that appear to be supportive from outside. The key effective way to fight the social barriers is through generating awareness. The continuing research, discussion and all forms of engagement must continue among local and wider communities until majority of the people are conscious of the existence of prejudices to provide a gender-neutral learning space for women in computing.

The HCI, CSCW and ICTD communities should stand together in ensuring adequate representation, valuation and support for women in computing anywhere in the world. The problem is not a Bangladeshi or South Asian problem, it is a global problem and requires a louder voice raised.

8 LESSONS LEARNED AND WAY FORWARD

Our study, situated at the intersection of fictional inquiry, feminist HCI, and NLP, yielded several key lessons about participant engagement, methodological innovation, and implications for the design of culturally sensitive NLP tools.

8.1 Applying Fictional Inquiry for Participatory Policy Design

In our study we used fictional inquiry as a pathway for policy imagining and value articulation. Through letters addressed to fictional characters like Oporajita (a symbolic feminist figure) and M3 (a patriarchal persona), students engaged in emotionally rich, speculative storytelling that surfaced deeply personal reflections about institutional, infrastructural, and societal barriers to their well-being in computing education. The indirect nature of fictional inquiry facilitated honest disclosures that participants may have otherwise withheld, especially in institutional contexts where power imbalances exist. For example, while challenges like having only one internet router for thousands of students in a residential hall may be widely known, they are rarely expressed in formal settings. As insiders within the institutional context, we were acutely aware of how our presence could influence participant responses. Fictional inquiry helped mitigate this by shifting the locus of response away from the researchers and toward imagined futures. For example, students were more comfortable suggesting psychological evaluations for faculty through fictional letters than they would have been in direct interviews.

Fictional inquiry thus served as a powerful means of surfacing “unspeakable” or unspoken concerns, and enabled students to articulate their ideal futures, both for themselves and the future generations. Though our focus was the education sector, this methodology has clear potential for domains such as labor rights (e.g., garment workers), urban planning, and public health, sectors where direct critique is often inhibited by power relations. Thus, we believe that fictional inquiry can be further leveraged in many ways as follows:

- **Participatory policy design:** especially when working with marginalized communities where sharing will not be limited through indirections.
- **Discussion on difficult topics:** exploration on socially or emotionally challenging concepts where there is a fear of being judged or labelled easily.
- **Engagement of minors:** in many cases the voices of young ones are important, and this method can open up non-personal sharing through the indirection of FI.

8.2 Insights from Co-Design Workshops and Student Reflections

Our co-design workshops, conducted in different regions, affirmed the value of including participants with varying levels of academic experience. While first-year students often shared optimistic visions, they also exhibited limited awareness of structural constraints, potentially leading to more idealistic or far-fetched solutions. However, we adapted our participant recruitment to include more senior students whose lived experiences brought sharper critique and more grounded recommendations. This iterative adjustment underscores the importance of sampling diversity when designing participatory futures. Although many proposed idealistic interventions (e.g., smart bands for anxiety tracking), they also acknowledged implementation barriers and trade-offs. Because participants were already embedded in the systems they were critiquing, they naturally accounted for contextual constraints, requiring little external framing from researchers. Their responses reflected both realism and resilience, contributing to a set of policy and tool design ideas rooted in lived experience.

8.3 Reflections on NLP Integration and Model Limitations

As we applied NLP tools to analyze students' letters, we encountered different types of bias: societal bias reflected in student narratives and model bias evident in model misclassifications, particularly in culturally specific contexts. These biases represent epistemic concerns, questions about whose knowledge is valued and what counts as valid understanding, where models may systematically fail to recognize or validate the emotional expressions and cultural contexts of Bangladeshi women students, or succeed, depending on the capability of the models. This highlights that NLP tools are not neutral technologies but embody particular ways of knowing that may exclude or misrepresent marginalized communities' lived experiences. To mitigate these issues, we approached NLP tools reflexively. First, we conducted a full qualitative analysis before engaging in any NLP-based modeling. We then compared NLP outputs with human annotations by culturally

embedded researchers. For example, sentiment analysis revealed that participants expressed more logical, fact-based reasoning when writing to M3, and shared more affective, vulnerable thoughts with Oporajita. As we engaged with participants from both public and private universities across different regions, we were reminded of the importance of including diverse perspectives within our analysis, aligning with our commitment to a collaborative and equitable research process. These insights emerged not from the models alone, but from our triangulated human-machine interpretive process.

Furthermore, we identified failure cases and documented specific examples where NLP models misrepresented or overlooked emotional nuance. Our analysis suggests that open-source models, where pretraining datasets can be scrutinized, are preferable for feminist HCI research in the Global South. Current auditing tools for pretraining data are insufficiently transparent, limiting critical analysis of bias propagation. Grounded in that knowledge, we thus call for the development of culturally sensitive NLP tools, particularly for South Asian languages and social contexts. Building on work in data feminism and feminist NLP, we advocate for community-led data curation and model evaluation practices that better represent gendered experiences from underrepresented regions. We also emphasize several methodological takeaways:

- **Unit of Analysis Flexibility:** NLP tools should support analysis at varying levels of granularity (e.g., quote, paragraph, or full document). For example, in our case, quotes were the primary unit, but aggregating quotes into “letter-level” sentiment scores provided additional insight.
- **Transparent Tool Selection:** Tool choice was based on availability, pricing, and alignment with our dataset. This time, we chose open-source tools which we could use from Hugging Face Repository at no-cost which showed high scores on their evaluation set, along with having the pretraining data publicly available. Future work would benefit from clearer commentary and documentation on tool limitations, particularly when used in marginalized contexts.
- **Pretraining data audit:** We emphasize the importance of pretraining data auditing tools. While there are many data and model repositories worldwide, researchers face difficulty to identify which datasets were used to train which models along with the preprocessing steps.
- **Bias-Aware Frameworks:** We used a bias-aware framework for examining power relations in NLP tools to examine our data, however there was no way to address the power imbalances even after examining. There is an urgent need for research into tools that not only detect bias but offer pathways for correction or contextual feedback responses to model developers for systematic errors.

Our reflections point to the need for a more formalized pipeline for integrating feminist NLP into participatory feminist HCI research. Such a pipeline should support context-aware annotation, multilingual processing and gender bias analysis tools. We complement existing works in these contexts, by grounding our pipeline in lived, culturally specific narratives and highlighting the adaptation to South Asian (Bangladesh) context. Ultimately, this reflection reinforces the importance of researcher positionality and reflexivity in shaping not only the research outcomes but also the ethical responsibility of advocating for inclusive and contextually relevant technological solutions.

9 LIMITATIONS

This work considers tertiary level female students only without considering the perspectives of male peers or other stakeholders around women in computing. We wanted to focus entirely on the student perspective in-depth and hope to consider the other views with same rigor in future.

10 CONCLUSION

Grounded in the Bangladeshi context, this study highlights how Fictional Inquiry, and co-design can surface the emotional and structural barriers faced by Women in Computing (WiC). Our study offers future female computing professionals a platform to voice the challenges they have faced and anticipate in their academic

and professional journeys. Their solution ideas, while at an early stage, offer valuable starting points for future human-centered design and policy efforts. As we invited participants to express themselves through fictional letters and collaborative design, we were able to uncover perspectives often overlooked in traditional research. Moreover, with a combination of NLP and qualitative methods, we captured emotional depth while remaining mindful of the limitations of existing tools. We hope that our study will be a strong contribution to the careers and lives of Bangladeshi female computer science students and contribute to broader efforts to amplify WiC voices while adding to the growing momentum toward making computing education more inclusive and responsive to the realities of women in the field.

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11 APPENDIX

A. LETTERS USED IN FICTIONAL INQUIRY

Letters named- Oporajita and M3 used during the Fictional Inquire session. The upper side letter is from Oporajita representing a supporting persona and the lower side letter shows the persona of M3 who uses societal judgements on women's ability.

Hello

I am a member of Begum Rokeya's family. Today, I come before you to know more about you, your struggle and challenges.

My ancestor, Begum Rokeya envisioned a world where women and men have equal rights and opportunities. She raised her voice against injustice, prejudice and bigotry that deprived women of their right to education and financial independence.

During her time, she went above and beyond to generate awareness and build confidence among women. Her initiative 'Anjuman-e-Khawateen-e-Islam' was the forefront of the fight for women's education and employment. Begum Rokeya had laid the foundation for women's education back in her time. As I understand, now is the age of science and technology. The digital revolution has added new dimensions to education and work. It made knowledge on Science Technology Engineering and Mathematics (STEM) integral to progress.

As her descendent, I want to continue her mission and lead the way forward to the world once envisioned. I wish to know how you are doing in this new era. Do you still face challenges in your education? What are the injustices you fight every day? How are you coping with the new dimension in education and work?

Yours

Oporajita

Figure A1 Letters used in Fictional Inquire Session: A Letter from Oporajita

Hello all

I am here to listen to you and solve your problems. [Which problem?] I have a soft corner to help others. I am happy to see few women here in this team, I hope you will be different and passionate about your job.

I want to spend my money to solve problems – but do not want to waste it. So, please make sure you are serious about this problem of engaging more women in computing.

I am an executive in a multinational corporation in Bangladesh. You know my brand, it is one of the best company in the country. So if you can convince me about a problem, I can spend resources and connections you will need. You know my family – I'm sure.

But honestly, I think the problem here is about women's ability – they are not as sharp as men. I have seen it in many cases – for computing, men have a natural talent. Boys do so good. But when it comes to women, its just about looks and other things – they are never serious about job. I don't think women are as serious as men in this sector.

I am happy that you want to find out about women's barriers, but you will not find many. Mostly they are busy with themselves – they go to parlors, go to shopping and weddings – how will they focus on career? I have seen many like this in my family and among my friends. They are not interested to work. Many of them are quite capable but they have not pursued careers. You can be angry at me hearing the truth, but you will see them in your families too.

People make up stuff all the time to get money from me. So you have to convince me. As I promised, if you can give me the examples, cases and places that needs improvement, I will help.

Thanks

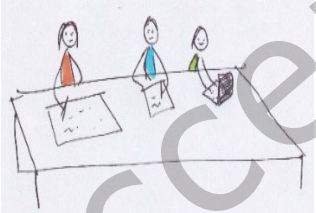
Mustakim Mohammad Monsur (M3)

Figure A2 Letters used in Fictional Inquire Session: Letter from M3

B. PICTORIAL REPRESENTATION OF WORKSHOP STEPS

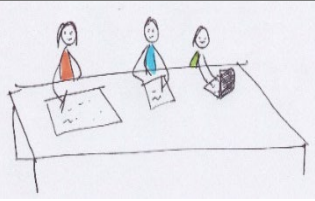
Fictional Inquiry Steps are represented through hand drawn sketches by researchers conducting the sessions to clarify the process visually as presented in Table B1.

Table B1: FI steps with pictorial representation

Step1	Step 2	Step 3
		
Receiving individual letter from Aparajita	Participants responds to individual letter	Group Discussion on challenges participants faced and shared in letters

Here the steps of co-design (CD) are presented using a pictorial representation in Table B2 that includes three major steps.

Table B2: CD steps with pictorial representation

Step1	Step 2	Step 3
		
Group discussion on major challenges identified from the letter responses in Fictional Inquiry workshops	Design concept and exploration by individuals working together in the same space.	Group Discussion with conclusion on solution approaches

C. EMOTION FREQUENCY ACROSS LOCATION

In the following heatmap of average emotion scores grouped by locations, we can see that neutrality and sadness are most prominent emotions. Other emotion scores are also fairly consistent over all locations. To construct the graph we average the emotion scores over M3 and Oporajita combined dataset and group by the locations

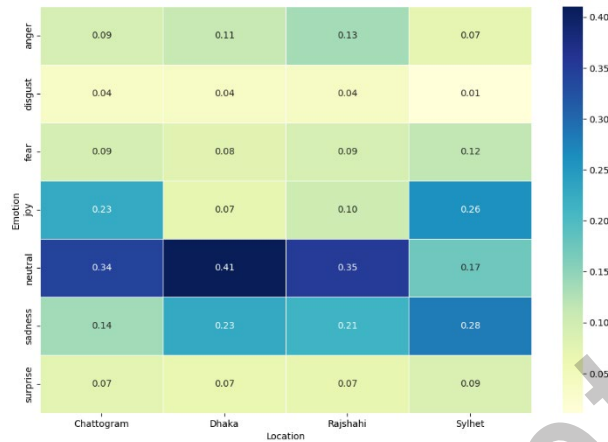


Figure C1: Combined location wise emotion analysis for M3 and Oporajita quotes

D. EMOTION FREQUENCY OF CHALLENGE SUB CATEGORY

When we investigate the emotions associated with each challenge sub-category in M3 vs Oporajita quotations as presented in Figure D1, we can see that academic challenges came up more in Oporajita quotes along with disgust and fear related emotions. The bias category shows disgust and fear for both Oporajita and M3, but students show anger only to Oporajita. Students talked about their transportation related fears to Oporajita more too.

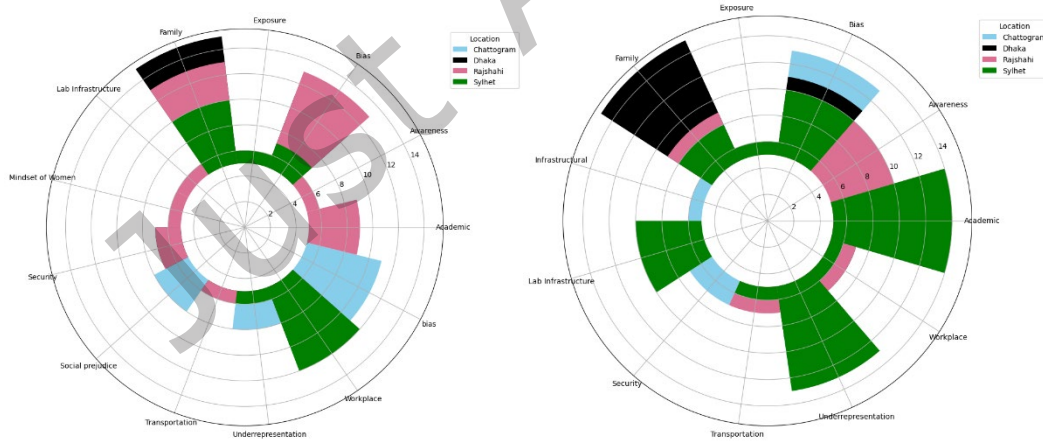


Figure D1. Frequency of Challenges Sub-Category for M3 and Oporajita

E. AVERAGE SENTIMENT PER WORKSHOP

The mean sentiment scores across Oporajita, M3 and design workshop shows how neutral, negative and positive score values varied across the workshops as presented in Figure E1.

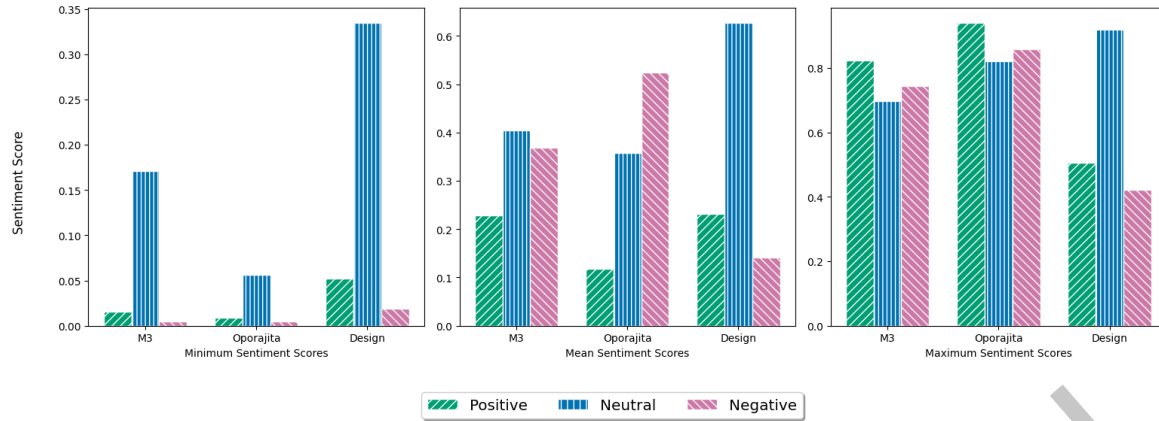


Figure E1. Sentiment score average, minimum and maximum across various letter response

F. MODEL EVALUATION GUIDELINES

We followed these guidelines when conducting human annotation in consideration of potential errors.

1. Translation changed the meaning of a sentence (slangs were removed, insults toned down, or emotion softened during automatic translation).
2. The NLP model misunderstood or erased participant intent (mislabeling a hopeful statement as neutral, mistranslating gendered expressions, or skipping culturally specific terms).
3. The model failed to pick up sarcasm or humor (someone said something playful or ironic and the model interpreted it literally)
4. The model failed to detect the correct sentiment/emotion (model marking something as “positive” when it’s clearly “negative” or marking something as “neutral” when its frustration or sadness).
5. There was disagreement among annotators or the team (if multiple reviewers had different interpretations of emotion, intent, or meaning). This shows how subjectivity and positionality affect our work.